
CPE 324: Digital Signal Processing

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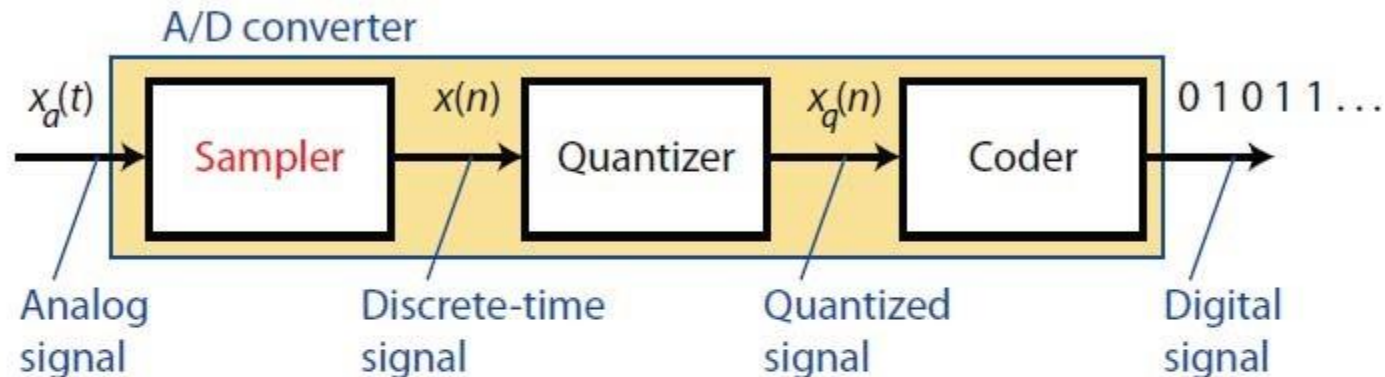


Week 10: **Sampling and Reconstruction of signals**

- ❑ **Sampling Process: Revisited**
- ❑ **Sampling and Interpolation Methods**

Sampling Process: Revisited

Analog-to-Digital Conversion



Sampling:

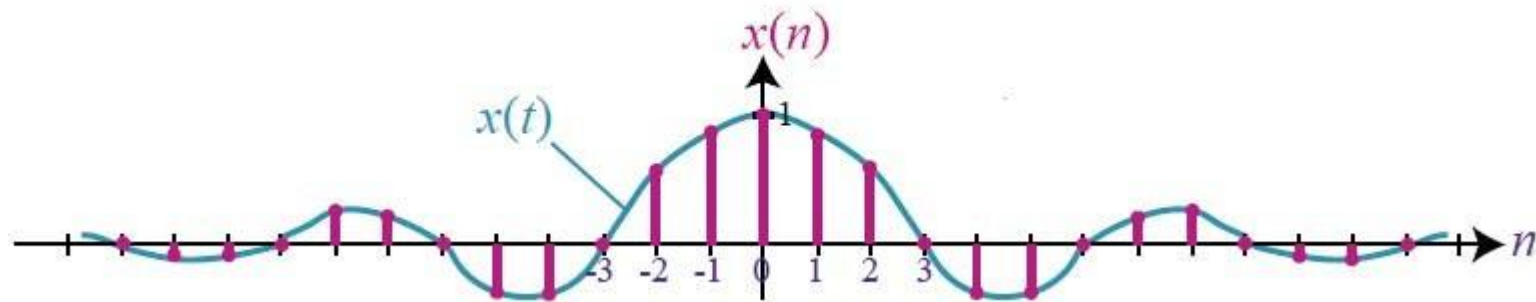
- ▶ conversion from cts-time to dst-time by taking "samples" at discrete time instants
- ▶ E.g., uniform sampling: $x(n) = x_a(nT)$ where T is the sampling period

Sampling Process: Revisited

Sampling Process

- ▶ To effectively reconstruct an analog signal from its samples, the sampling frequency $F_s = \frac{1}{T}$ must be selected to be "large enough".
- ▶ Sampling in the time-domain:

$$x(n) = x_a(nT), \quad -\infty < n < \infty$$



Sampling Process: Revisited

Sampling Process

- ▶ Time-Domain Sampling: frequency-domain perspective
 - ▶ The sampling frequency $F_s = \frac{1}{T}$ must be selected to be large enough such that the sampling processing does not cause any **loss of spectral information** (i.e., no aliasing).

- ▶ Recall CTFT and DTFT for **aperiodic $x_a(t)$ and $x(n)$** :

Only this class is considered as periodic signal do not exist in nature

$$\begin{aligned}x_a(t) &= \int_{-\infty}^{\infty} X_a(F) e^{j2\pi Ft} dF & x(n) &= \int_{-\frac{1}{2}}^{\frac{1}{2}} X(f) e^{j2\pi fn} df \\X_a(F) &= \int_{-\infty}^{\infty} x_a(t) e^{-j2\pi Ft} dt & X(f) &= \sum_{n=-\infty}^{\infty} x(n) e^{-j2\pi fn}\end{aligned}$$

Sampling Process: Revisited

Sampling Process

$$x(n) \equiv x_a(nT) = x_a\left(\frac{n}{F_s}\right) \implies \boxed{f = \frac{F}{F_s}}$$

$$\underbrace{\int_{-\frac{1}{2}}^{\frac{1}{2}} X(f) e^{j2\pi f n} df}_{LHS} = \underbrace{\int_{-\infty}^{\infty} X_a(F) e^{j2\pi F \frac{n}{F_s}} dF}_{RHS}$$

$$\begin{aligned} LHS &= \int_{-\frac{1}{2}}^{\frac{1}{2}} X(f) e^{j2\pi f n} df \quad \text{let } f = \frac{F}{F_s} \\ &= \int_{-\frac{F_s}{2}}^{\frac{F_s}{2}} X\left(\frac{F}{F_s}\right) e^{j2\pi \frac{F}{F_s} n} \frac{dF}{F_s} \\ &= \frac{1}{F_s} \int_{-\frac{F_s}{2}}^{\frac{F_s}{2}} X\left(\frac{F}{F_s}\right) e^{j2\pi \frac{F}{F_s} n} dF \end{aligned}$$

Sampling Process: Revisited

$$\begin{aligned} RHS &= \int_{-\infty}^{\infty} X_a(F) e^{j2\pi \frac{F}{F_s} n} dF \\ &= \sum_{k=-\infty}^{\infty} \int_{-\frac{F_s}{2} - kF_s}^{+\frac{F_s}{2} - kF_s} X_a(F) e^{j2\pi \frac{F}{F_s} n} dF \quad \text{let } F' = F + kF_s \\ &= \sum_{k=-\infty}^{\infty} \int_{-\frac{F_s}{2}}^{\frac{F_s}{2}} X_a(F' - kF_s) \underbrace{e^{j2\pi \frac{F' - kF_s}{F_s} n}}_{= e^{j2\pi \frac{F'}{F_s} n}} dF' \\ &= \int_{-\frac{F_s}{2}}^{\frac{F_s}{2}} \left[\sum_{k=-\infty}^{\infty} X_a(F - kF_s) \right] e^{j2\pi \frac{F}{F_s} n} dF \end{aligned}$$

Sampling Process: Revisited

Sampling Process

Since,

$$\begin{array}{c} \text{LHS} = \text{RHS} \\ \frac{1}{F_s} \int_{-\frac{F_s}{2}}^{\frac{F_s}{2}} X\left(\frac{F}{F_s}\right) e^{j2\pi \frac{F}{F_s} n} dF = \int_{-\frac{F_s}{2}}^{\frac{F_s}{2}} \left[\sum_{k=-\infty}^{\infty} X_a(F - kF_s) \right] e^{j2\pi \frac{F}{F_s} n} dF \end{array}$$

By inspection,

$$X\left(\frac{F}{F_s}\right) = F_s \sum_{k=-\infty}^{\infty} X_a(F - kF_s)$$

or letting $f = \frac{F}{F_s}$,

$$X(f) = F_s \sum_{k=-\infty}^{\infty} X_a(f \cdot F_s - kF_s) = F_s \sum_{k=-\infty}^{\infty} X_a((f - k)F_s)$$

FT of given $x(t)$ signal

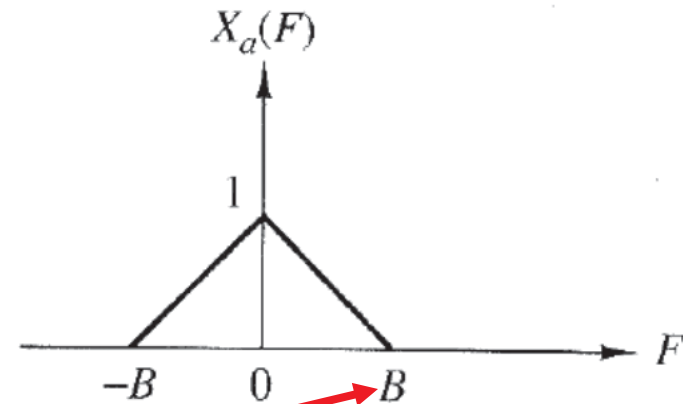
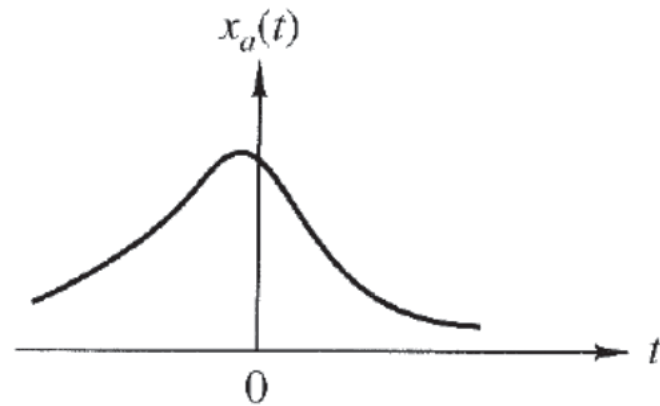
FT of the sampled version of $x(t)$

Sampling Process: Revisited

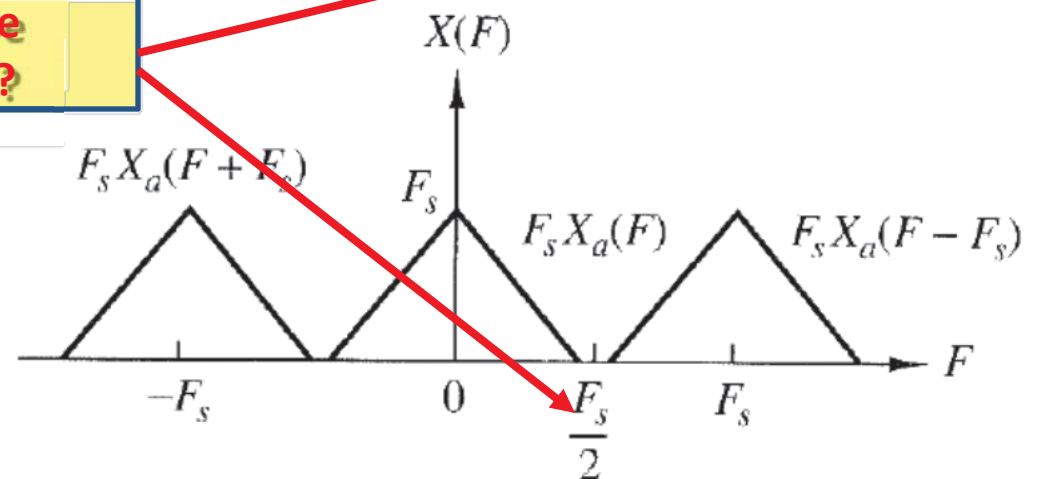
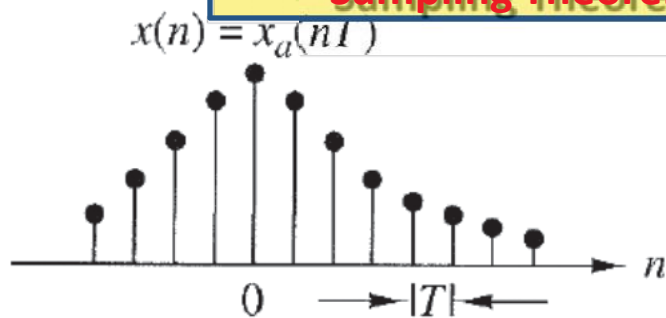
Sampling Process

- ▶ The spectrum of the sampled signal $x(n)$ is a scaled **periodic repetition** of spectrum of the original analog signal $x_a(t)$.
- ▶ The form of the periodic repetition guarantees that the signal $X(f)$ is periodic with fundamental period 1 or $X(\omega)$ is periodic with fundamental period 2π .

Sampling Process: Revisited



Guess what should be
Sampling Theorem ??



Scaled periodic repetition

Sampling Process: Revisited

Sampling Theorem

What about non-bandlimited signals? -- will deal later

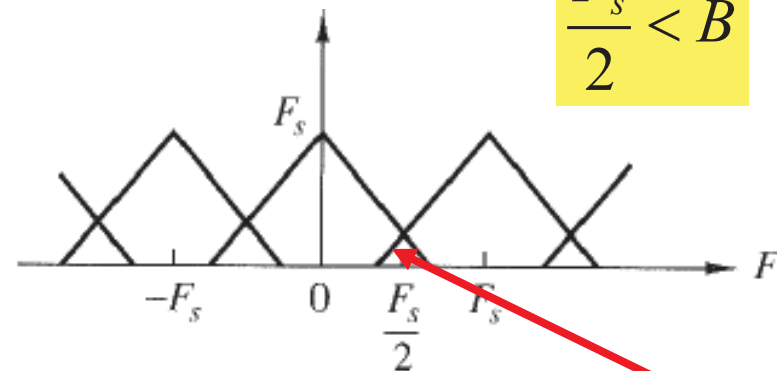
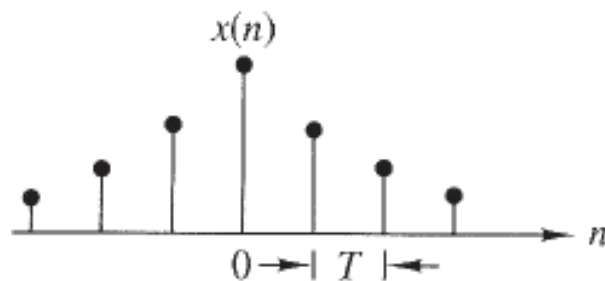
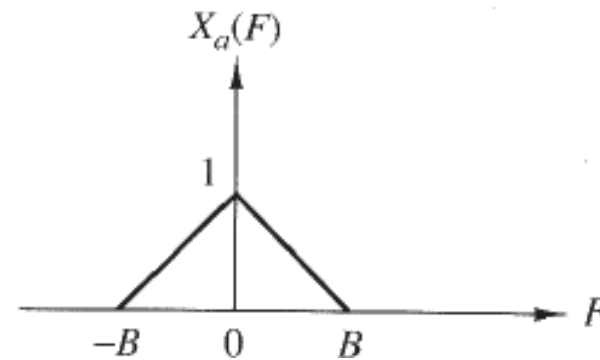
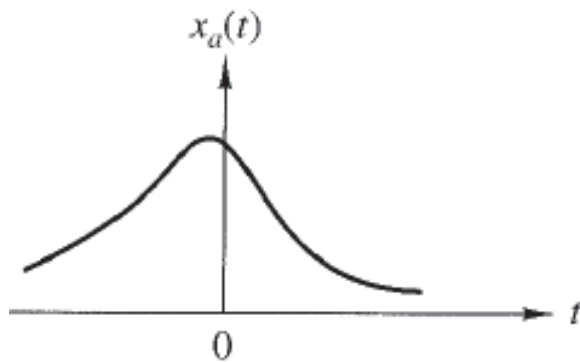
A **bandlimited** continuous-time signal, with highest frequency (bandwidth) B Hz, can be uniquely recovered from its samples provided that the sampling rate $F_s \geq 2B$ samples per second.

- Perfect reconstruction is possible via the ideal interpolation formula:

$$x_a(t) = \sum_{n=-\infty}^{\infty} \underbrace{x_a(nT)}_{=x(n)} \frac{\sin\left(\frac{\pi}{T}(t - nT)\right)}{\frac{\pi}{T}(t - nT)}$$

Sampling Process: Revisited

What happens in frequency domain if sampling theorem is not satisfied?

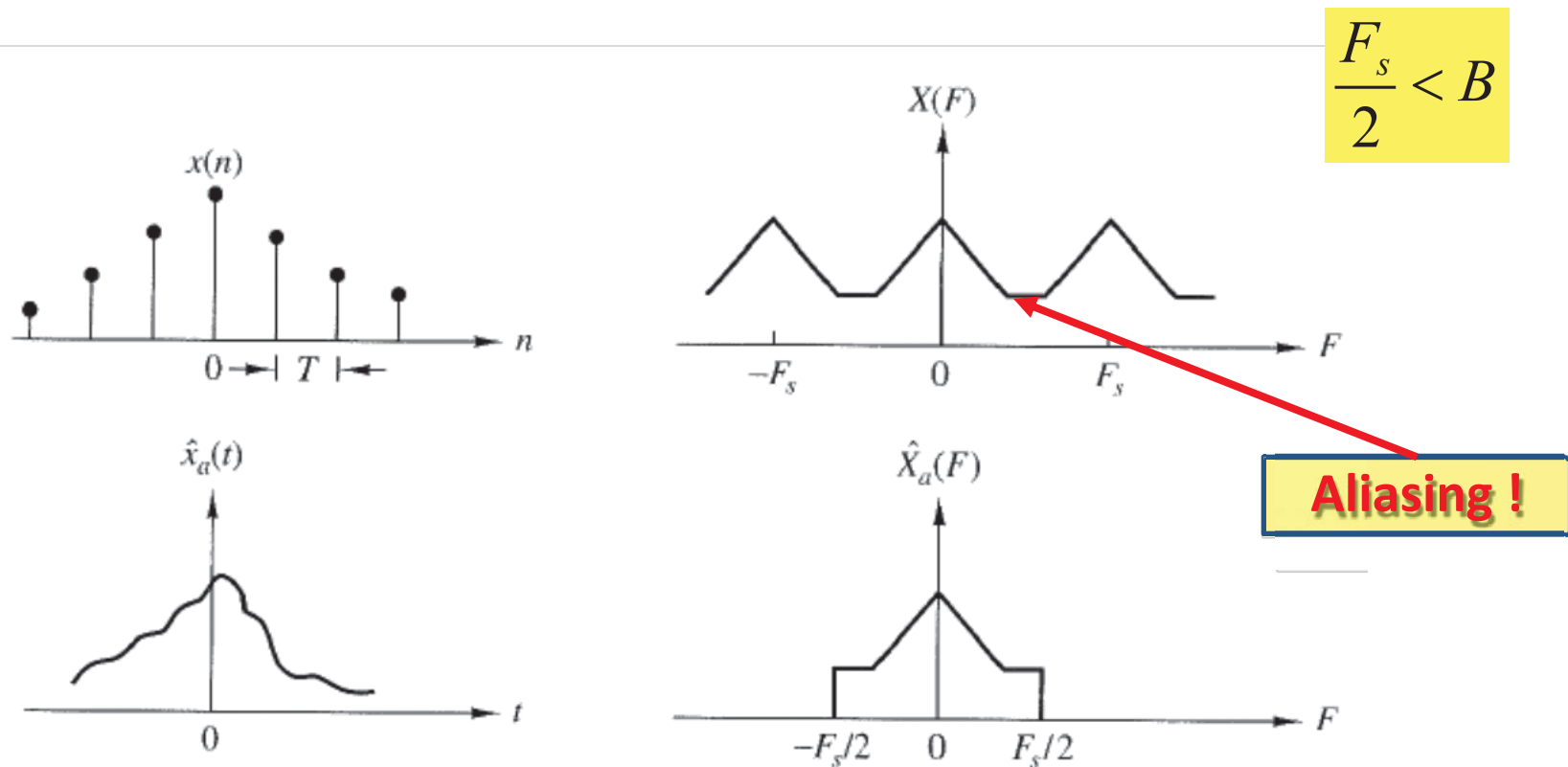


$$\frac{F_s}{2} < B$$

Aliasing !

Sampling Process: Revisited

What happens in frequency domain if sampling theorem is not satisfied?



Reconstructed signal is not same

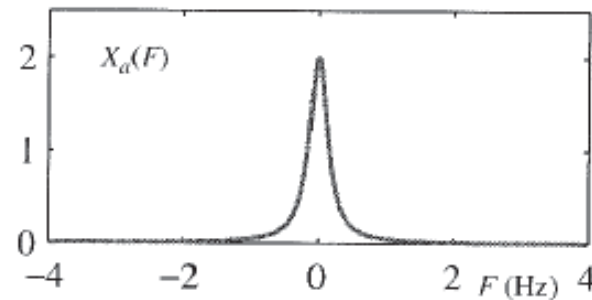
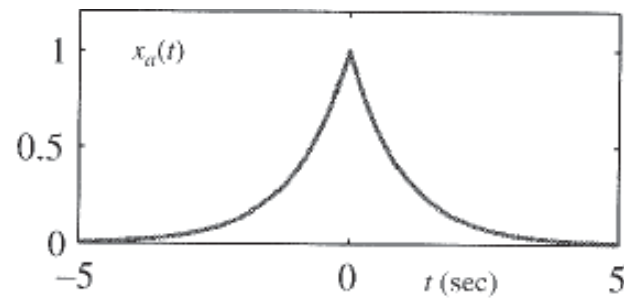
Sampling Process: Revisited

Non-bandlimited Signals:

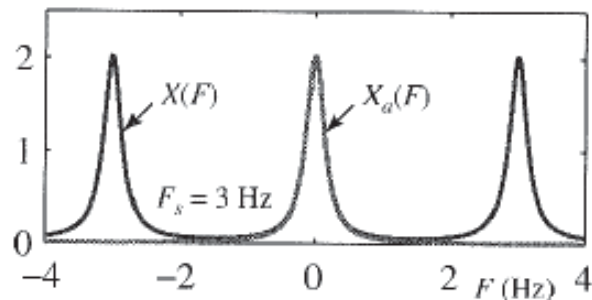
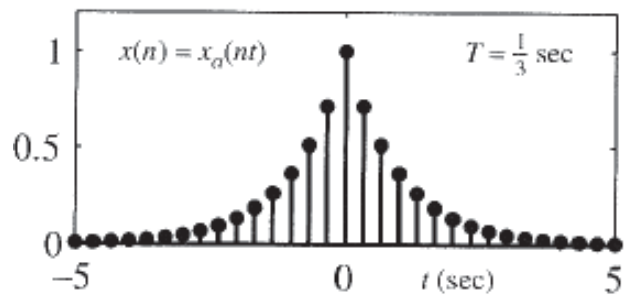
- ▶ Sampling and reconstruction of **nonbandlimited** signals results in **aliasing**.
- ▶ The degree of aliasing/quality of the reconstruction depends on the **sampling rate** in relation to the **decay of the analog signal spectrum**.
- ▶ Example:

$$x_a(t) = e^{-A|t|} \xleftrightarrow{\mathcal{F}} X_a(F) = \frac{2A}{A^2 + (2\pi F)^2}, \quad A > 0$$

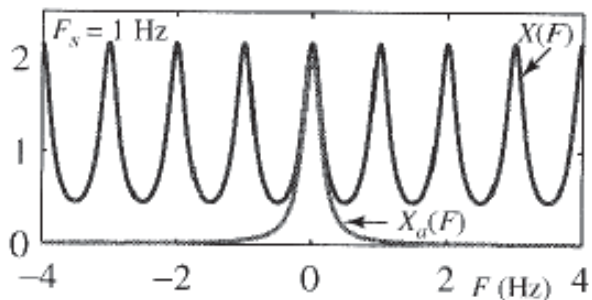
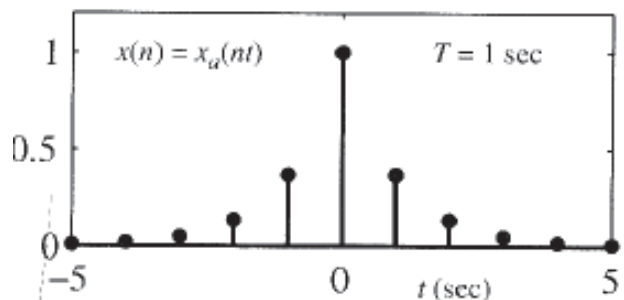
Sampling Process: Revisited



(a)



(b)



(c)

**Always
some
aliasing !**

Sampling Process: Revisited

Non-bandlimited Signals:

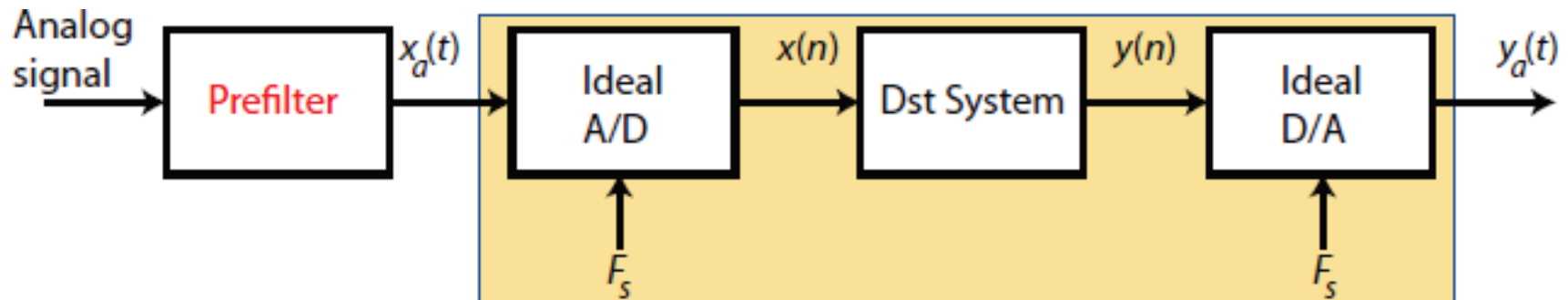
**Always some
aliasing !**

Solution ?

Anti-aliasing Filter

Sampling Process: Revisited

Anti-aliasing Filter



- ▶ ensures that bandwidth is limited to avoid aliasing or reduce subsequent computational requirements
- ▶ rejects additive noise in higher frequency ranges

Sampling Process: Revisited

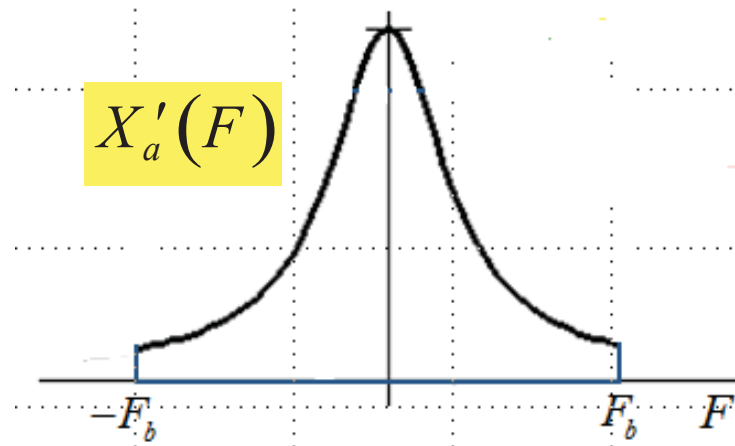
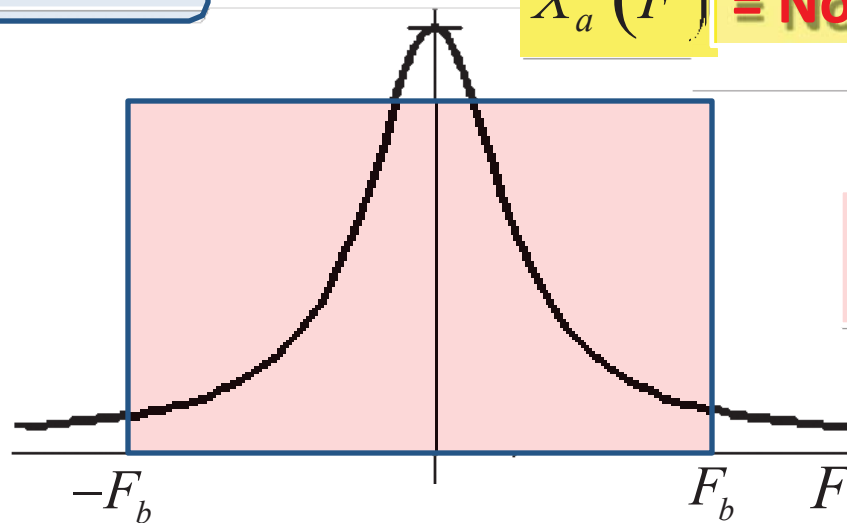
Anti-aliasing Filter

$X_a(F)$ = Non-bandlimited

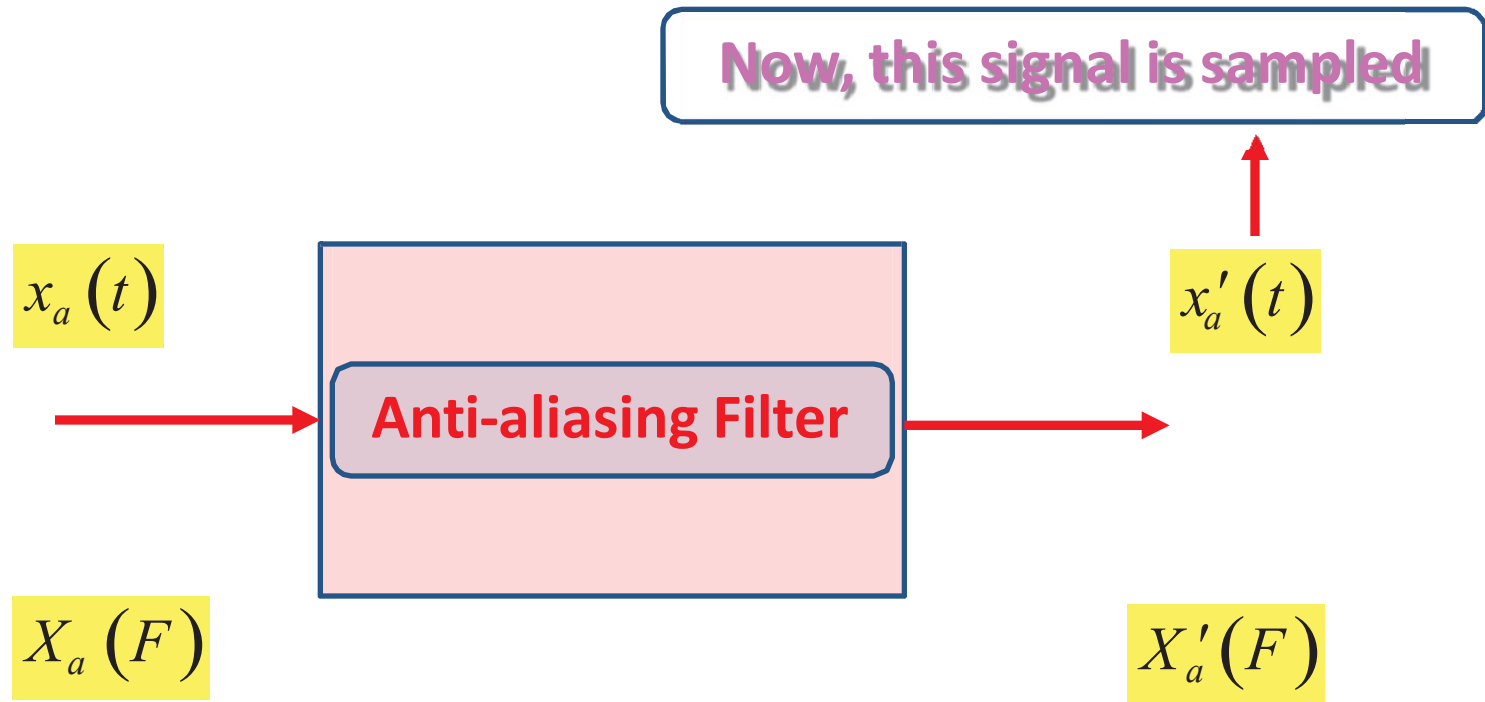
$h(n)$ for anti-aliasing filter
=
?

Anti-aliasing filter response
= Ideal low pass filter

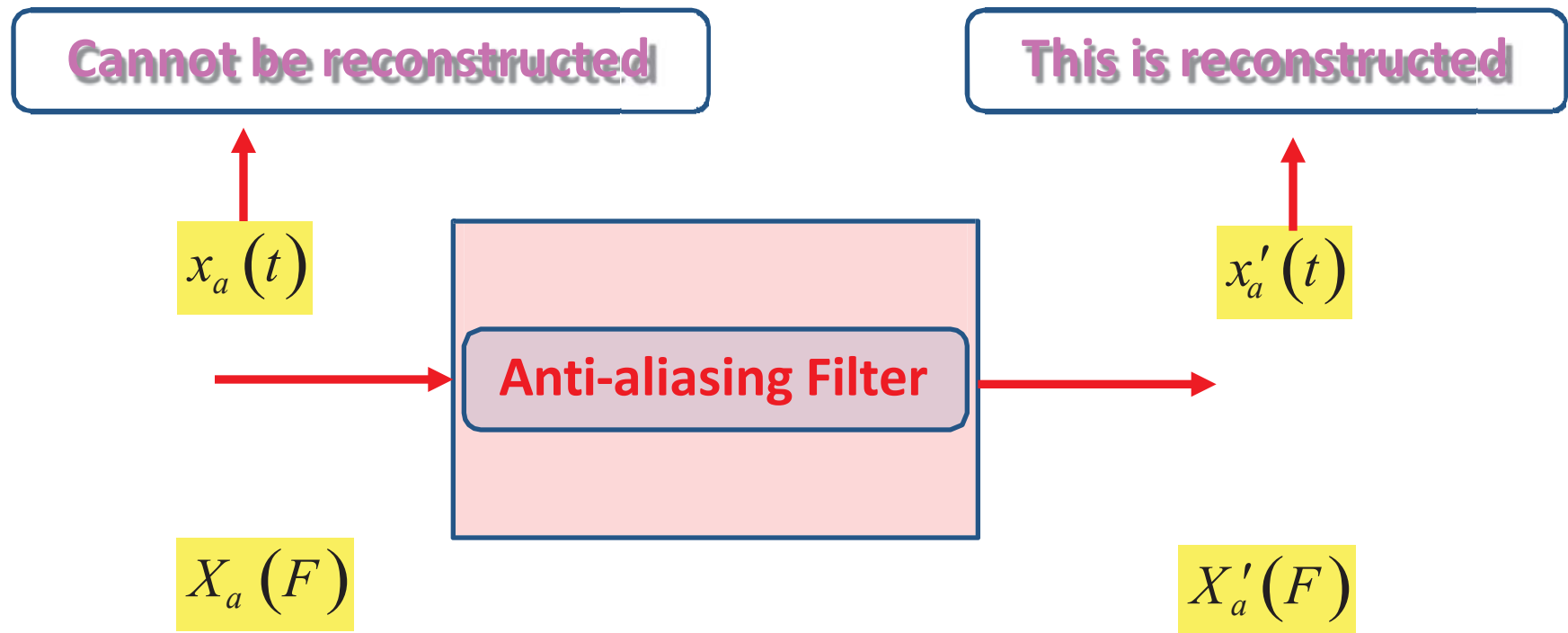
$$F_b \leq \frac{F_s}{2}$$



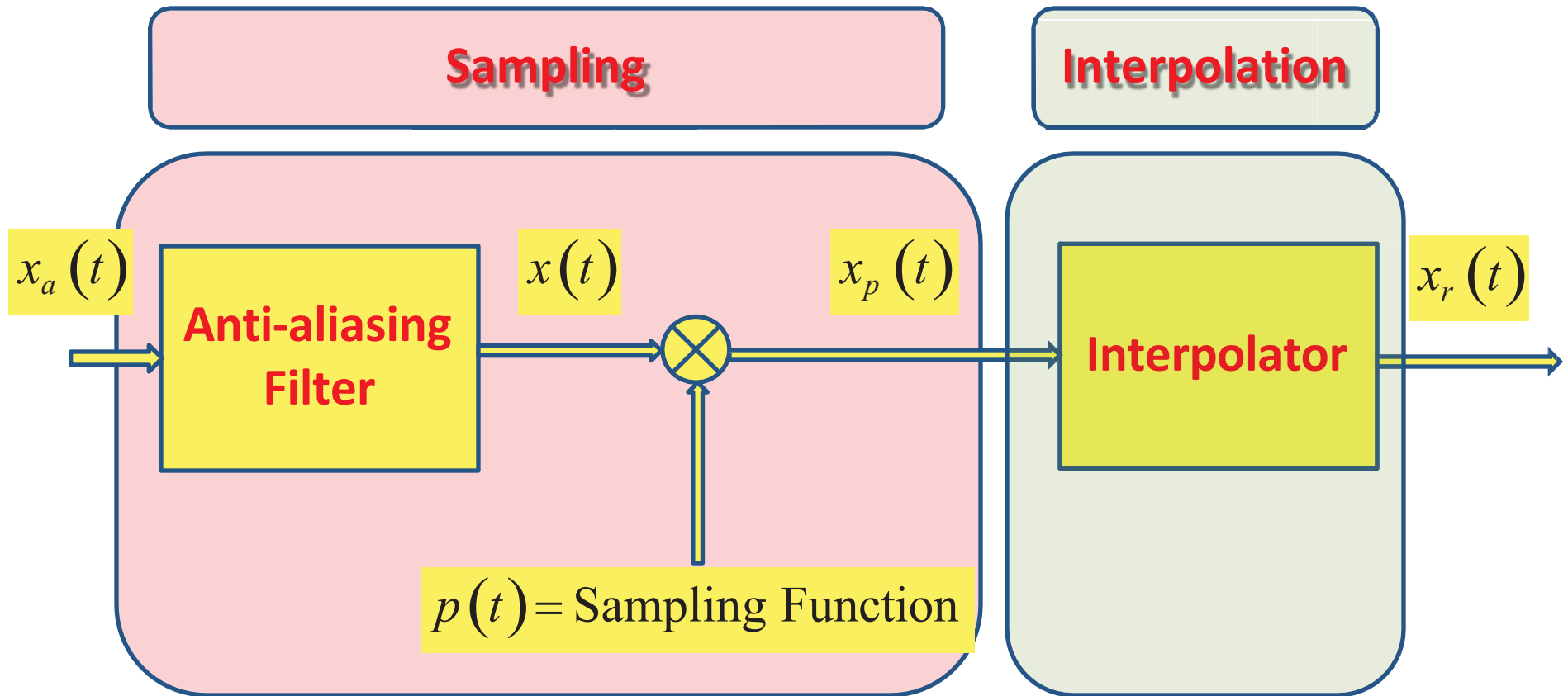
Sampling Process: Revisited



Sampling Process: Revisited



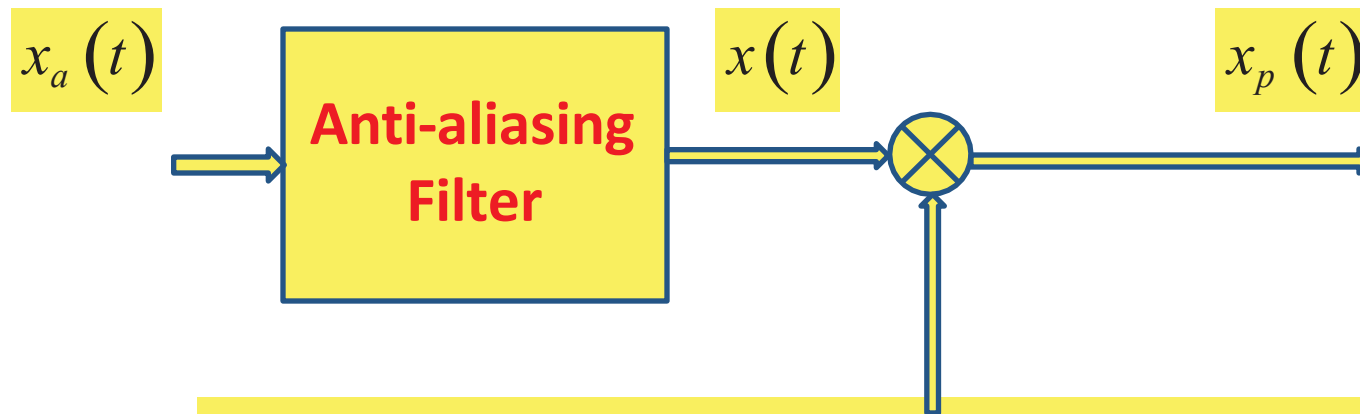
Sampling and Interpolation



Sampling and Interpolation

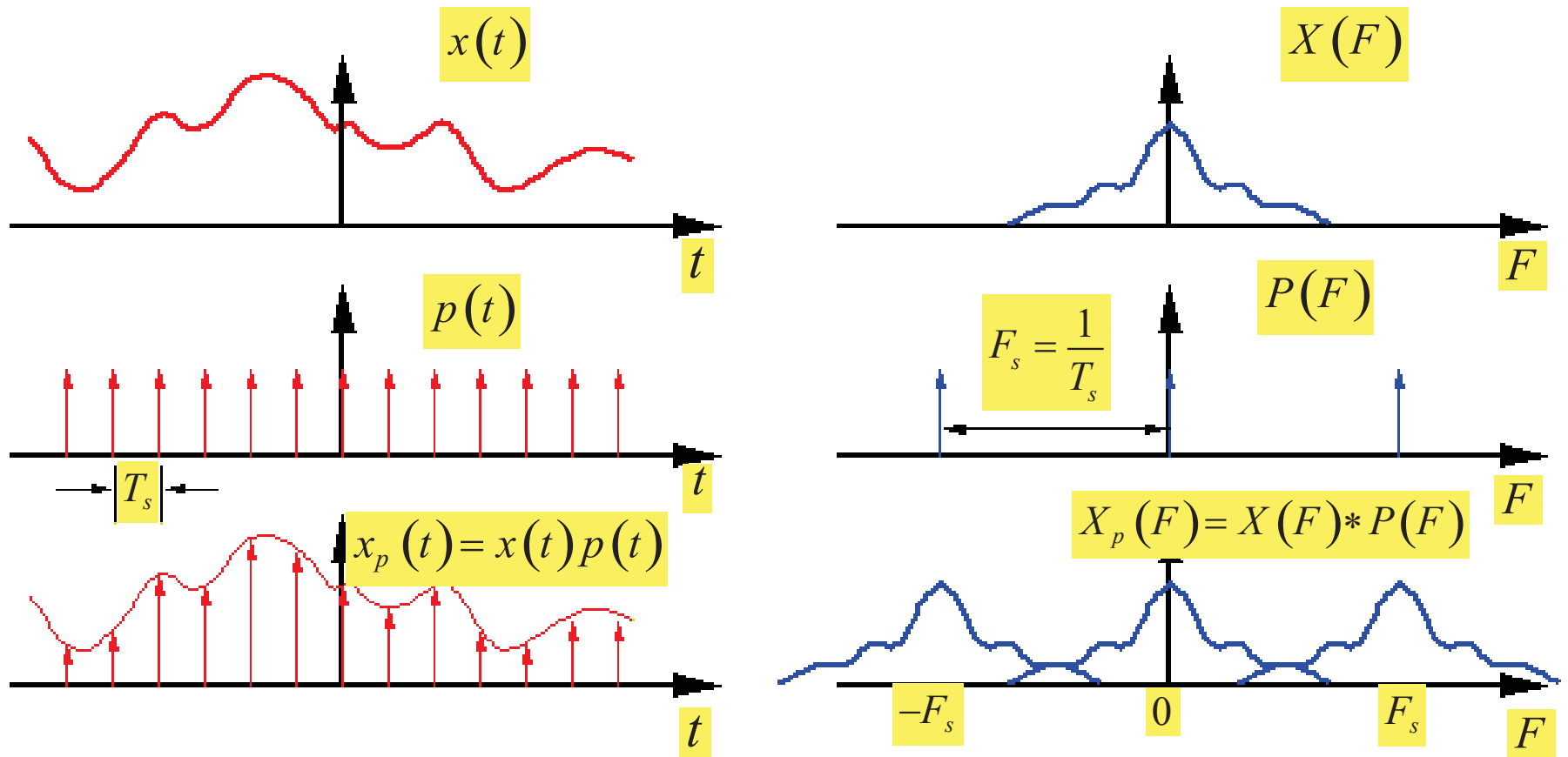
Ideal Sampling:

Impulse Train Sampling



$$p(t) = \text{Sampling Function} = \text{Impulse Train} = \sum_{n=-\infty}^{\infty} \delta(t - nT_s)$$

Sampling and Interpolation



Sampling and Interpolation

$$x(t) \Leftrightarrow X(F)$$

$$p(t) \Leftrightarrow P(F) = \frac{1}{T_s} \sum_{k=-\infty}^{\infty} \delta(F - kF_s)$$

CTFS of periodic impulse train

$$\begin{aligned} x_p(t) = x(t)p(t) &\Leftrightarrow X_p(F) = X(F) * P(F) = X(F) * \left[\frac{1}{T_s} \sum_{k=-\infty}^{\infty} \delta(F - kF_s) \right] \\ &= \frac{1}{T_s} \sum_{k=-\infty}^{\infty} X(F - kF_s) \end{aligned}$$

What's wrong with impulse-sampling? In practice, narrow and large-amplitude impulses are difficult to generate.

Sampling and Interpolation

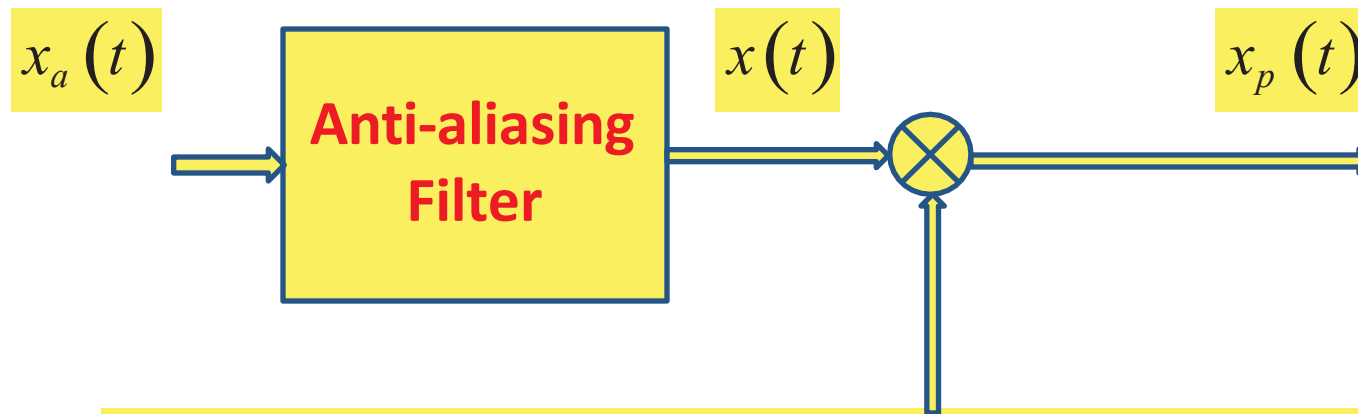
Ideal sampling produces a train of Dirac delta functions, and the weight of each delta function is proportional to the value of the sampler input at the time of sampling.

The output of the sampler when viewed in frequency domain is a set of equally spaced and scaled copies of $X(f)$. The spacing between copies is F_s and scaling factor is constant for all copies and is also equal to F_s .

What's wrong with impulse-sampling? In practice, narrow and large-amplitude impulses are difficult to generate.

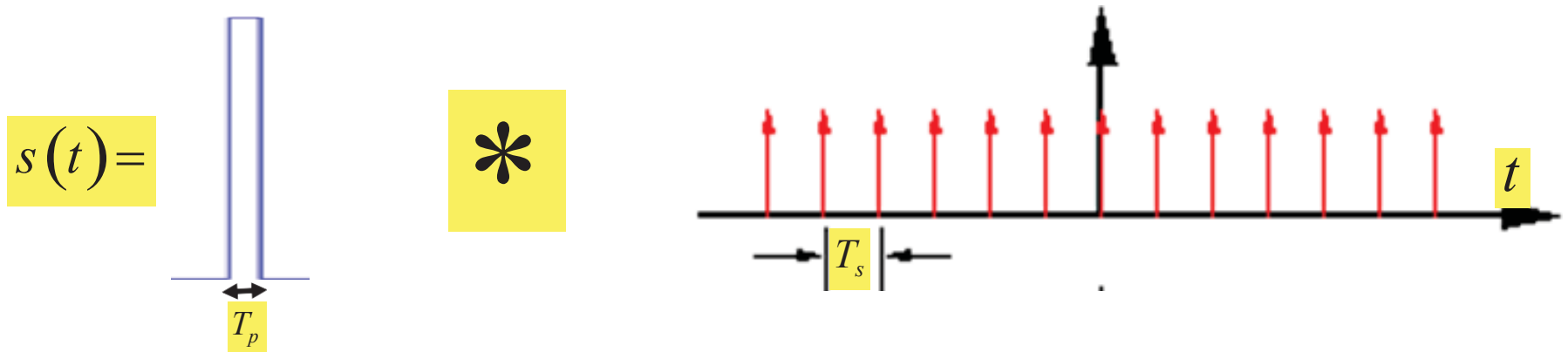
Sampling and Interpolation

Natural Sampling: Pulse amplitude modulation (PAM)



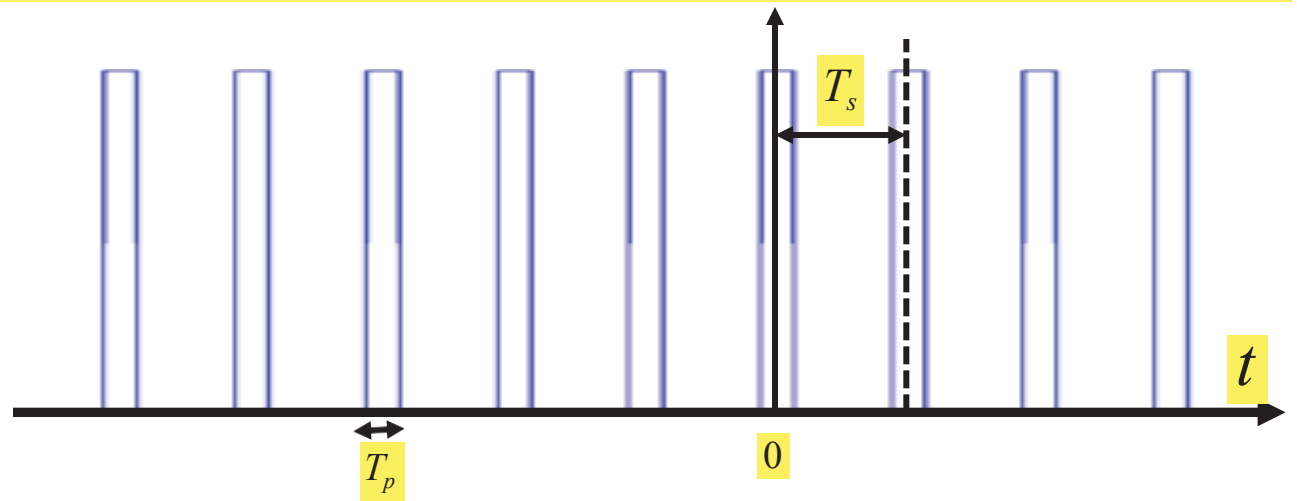
$$p(t) = \text{Sampling Function} = \text{Pulse Train} = \sum_{n=-\infty}^{\infty} s(t - nT_s) = s(t) * \left[\sum_{n=-\infty}^{\infty} \delta(t - nT_s) \right]$$

Sampling and Interpolation



=

$$p(t) = \text{Sampling Function} = \text{Pulse Train} = \sum_{n=-\infty}^{\infty} s(t - nT_s) = s(t) * \left[\sum_{n=-\infty}^{\infty} \delta(t - nT_s) \right]$$



Sampling and Interpolation

$$x(t) \Leftrightarrow X(F)$$

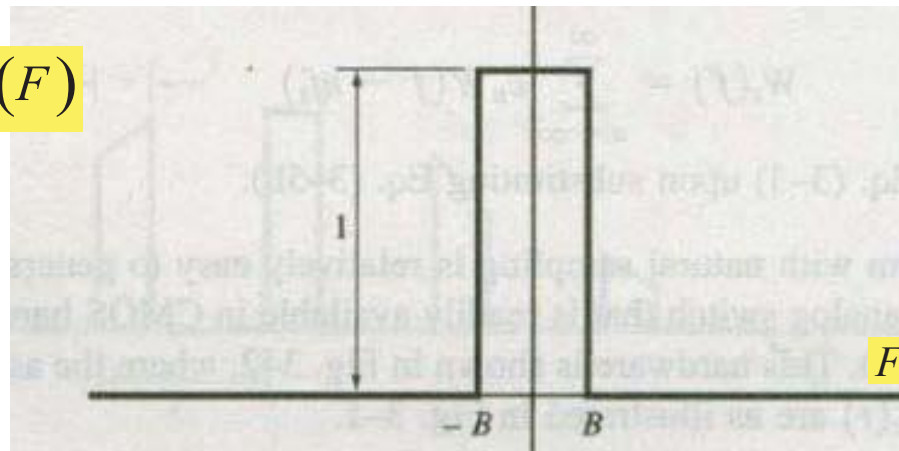
$$s(t) \Leftrightarrow S(F) = AT_p \frac{\sin(\pi FT_p)}{\pi FT_p}$$

$$p(t) = s(t) * \sum_{k=-\infty}^{\infty} \delta(t - kT_s) \Leftrightarrow P(F) = S(F) \cdot \left[\frac{1}{T_s} \sum_{k=-\infty}^{\infty} \delta(F - kF_s) \right]$$

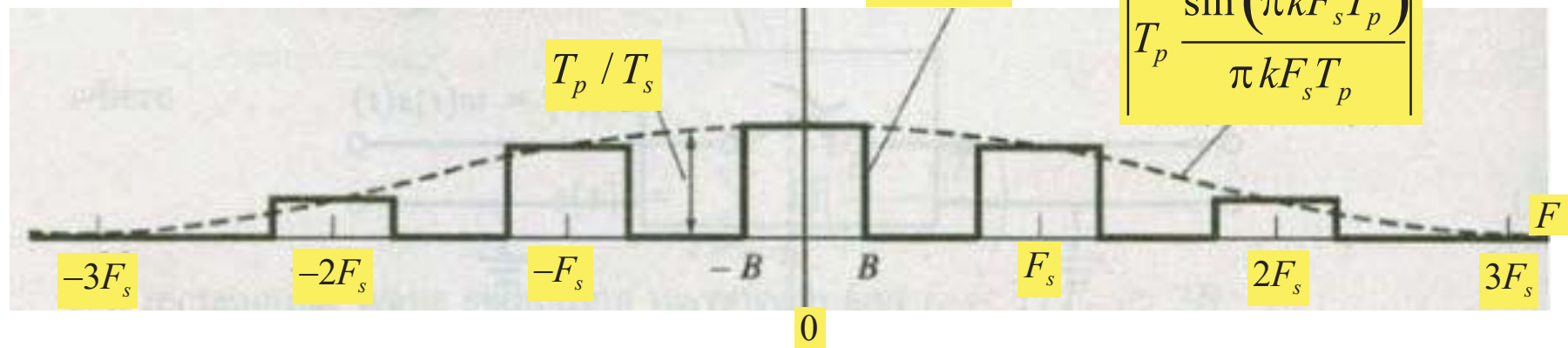
$$\begin{aligned} x_p(t) = x(t)p(t) &\Leftrightarrow X_p(F) = X(F) * P(F) = X(F) * \left\{ S(F) \left[\frac{1}{T_s} \sum_{k=-\infty}^{\infty} \delta(F - kF_s) \right] \right\} \\ &= X(F) * \left\{ \frac{1}{T_s} \sum_{k=-\infty}^{\infty} S(kF_s) \delta(F - kF_s) \right\} \\ &= \frac{1}{T_s} \sum_{k=-\infty}^{\infty} S(kF_s) X(F - kF_s) \end{aligned}$$

Sampling and Interpolation

$X(F)$

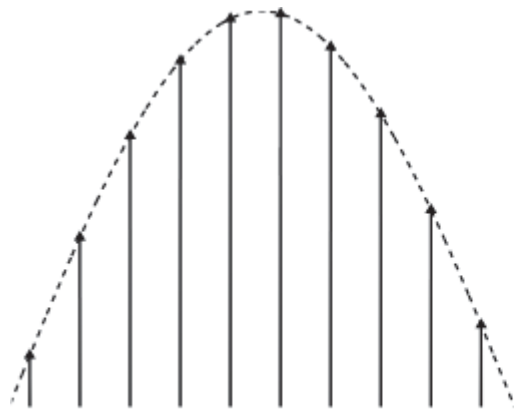


$X_p(F)$

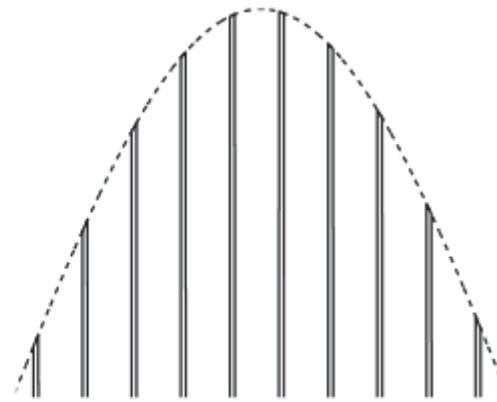


Sampling and Interpolation

Natural sampling is a practical method that approximates ideal sampling. The analog input is multiplied by a train of uniformly spaced, rectangular pulses. If the width of the pulses is much smaller than the spacing between pulses, then natural sampling may be regarded as an approximation of ideal sampling



Ideal Sampling

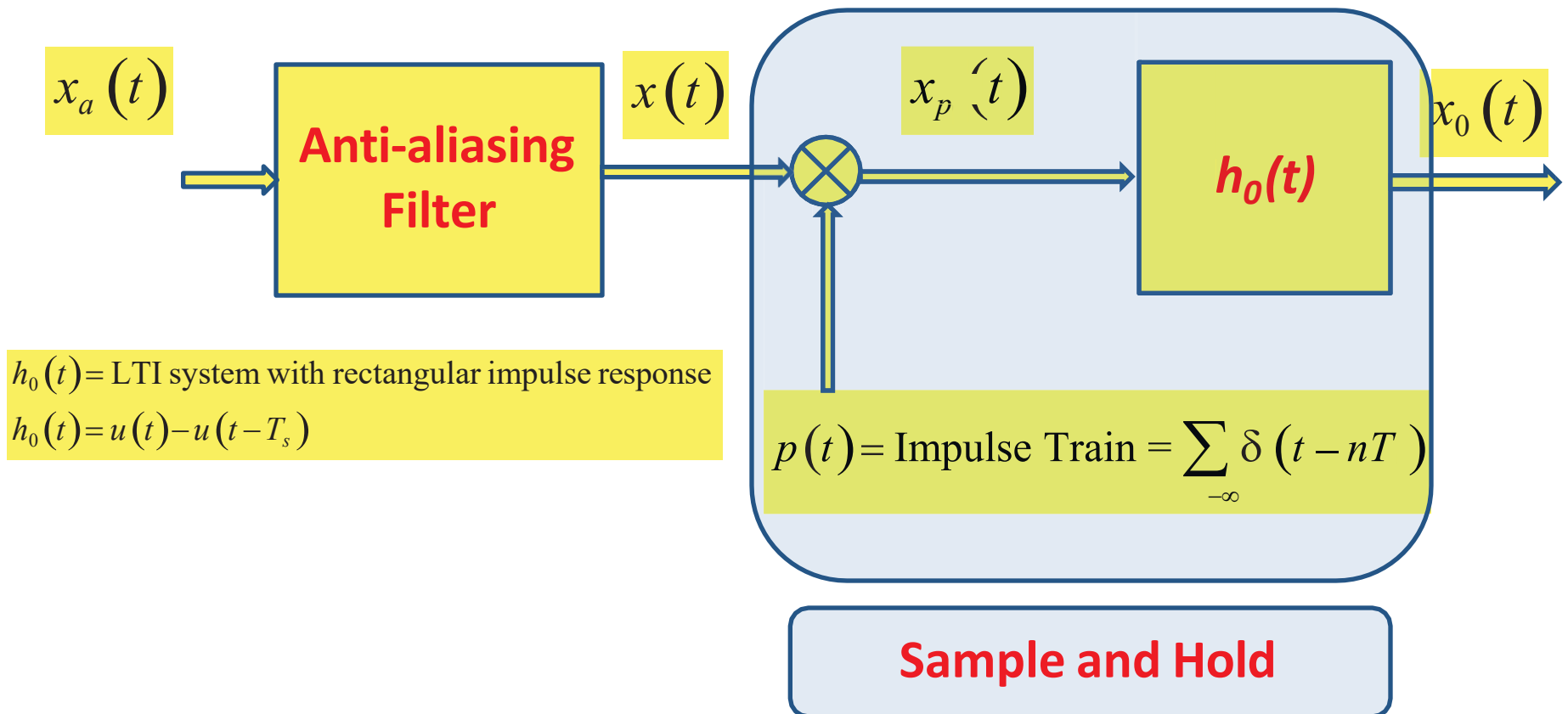


Natural Sampling

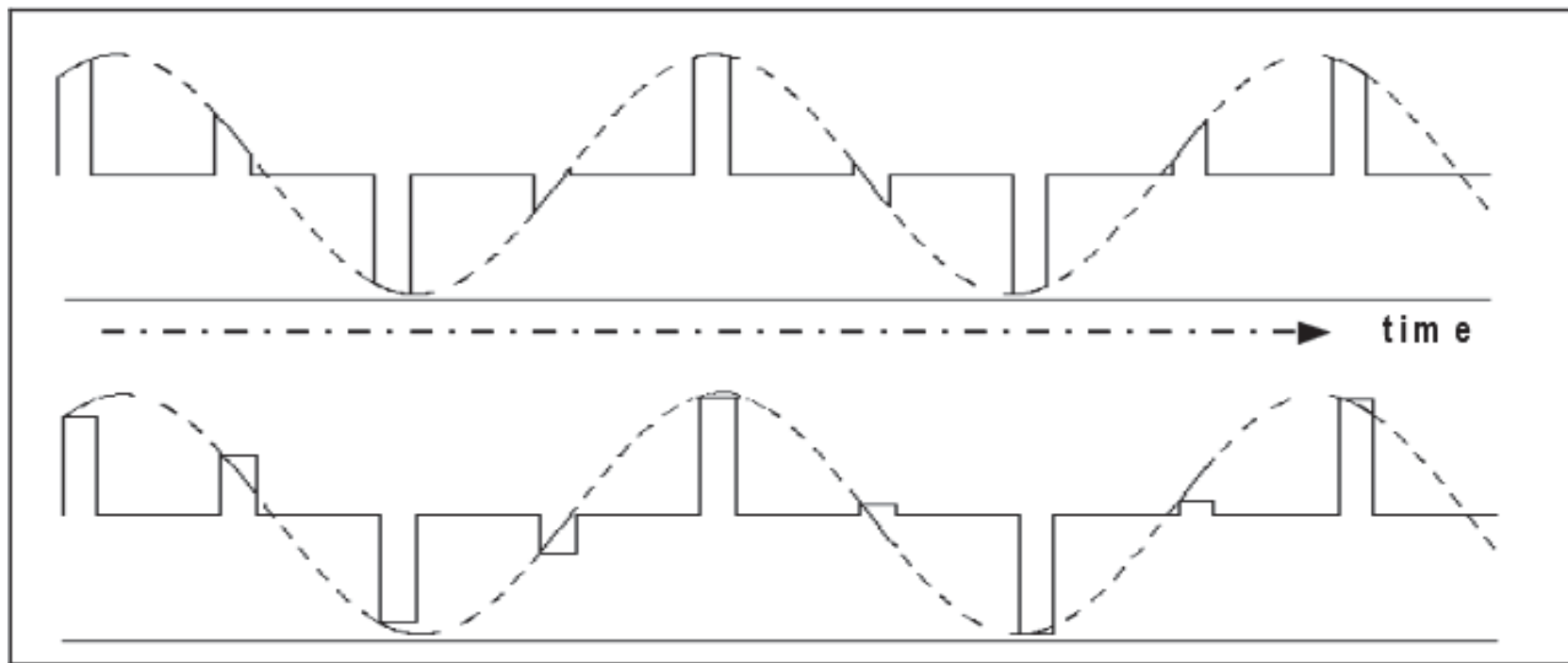
The output of the sampler when viewed in frequency domain is again a set of equally spaced and scaled copies of $X(f)$. The spacing between copies is F_s but now the scaling factor is not constant for all copies.

Sampling and Interpolation

Sample-and-hold or flat-top Sampling: Sampling with a zero-order hold



Sampling and Interpolation



natural sampling (above) and flat top (below)

Sampling and Interpolation

$$x(t) \Leftrightarrow X(F)$$

$$p(t) \Leftrightarrow P(F) = \frac{1}{T_s} \sum_{k=-\infty}^{\infty} \delta(F - kF_s)$$

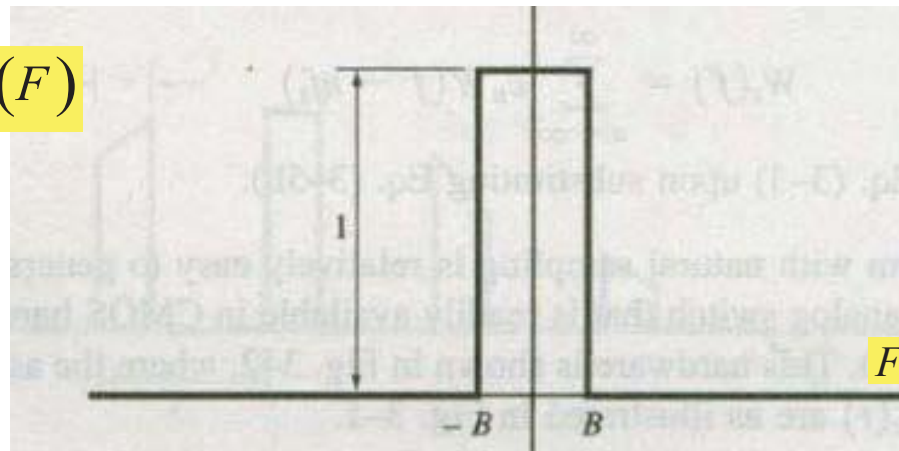
$$x_p(t) = x(t)p(t) \Leftrightarrow X_p(F) = \frac{1}{T_s} \sum_{k=-\infty}^{\infty} X(F - kF_s)$$

$$h_o(t) \Leftrightarrow H_o(F) = \frac{\sin(\pi FT_s)}{\pi F} e^{-j\pi FT_s}$$

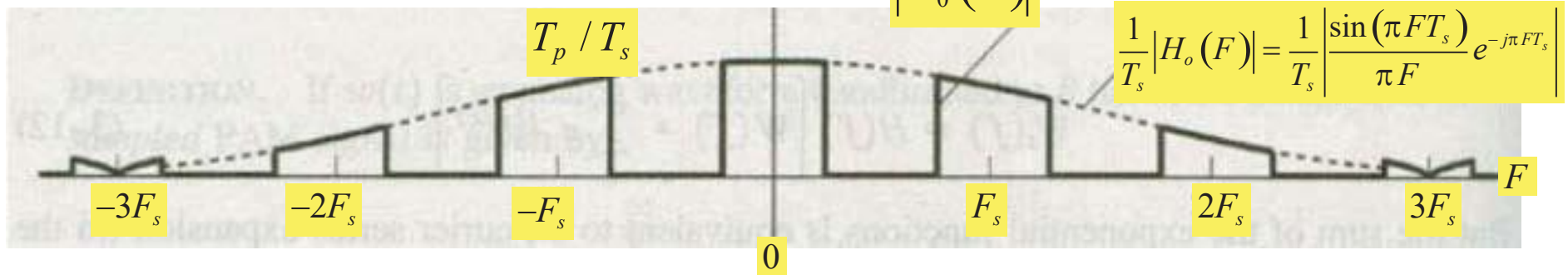
$$\begin{aligned} x_o(t) = x_p(t) * h_o(t) &\Leftrightarrow X_o(F) = X_p(F) \cdot H_o(F) \\ &= \frac{1}{T_s} H_o(F) \sum_{k=-\infty}^{\infty} X(F - kF_s) \end{aligned}$$

Sampling and Interpolation

$X(F)$



$|X_0(F)|$



Sampling and Interpolation

The most common sampler is the sample-and-hold device. The analog signal is sampled every T_s seconds. The sampler output holds steady at the last sample value, until a new sample is available. In this way, the sampler output is a staircase approximation to original analog signal.

In frequency domain, all copies of original analog signal spectrum are distorted, except for the case when bandwidth of original analog signal i.e. B is much smaller than F_s . In this case, only first copy i.e. copy corresponding to $F=0$, is not distorted but all other copies are distorted

Sampling and Interpolation

Interpolation is the estimation of the unknown, or the lost, samples of a signal using a weighted average of a number of known samples at the neighborhood points.

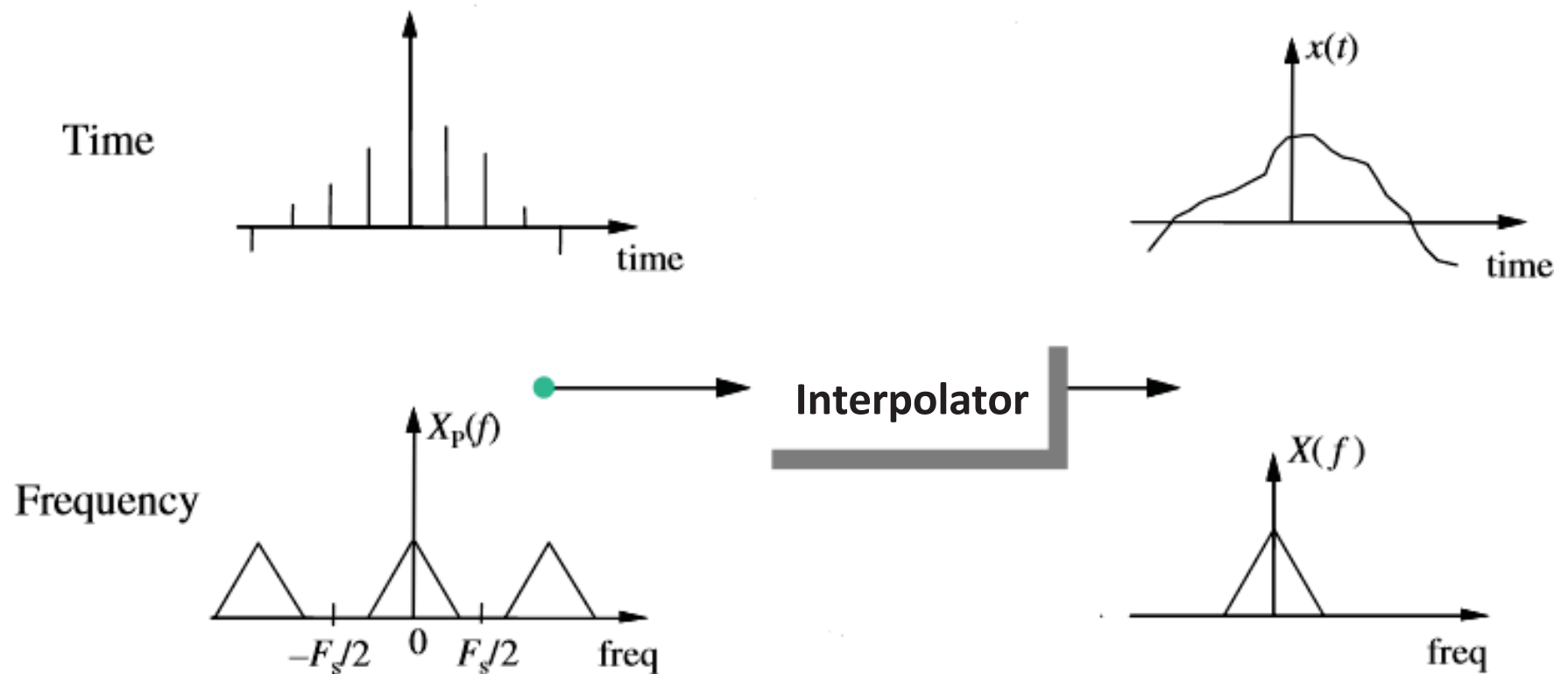
Reconstruction of analog
signal from its samples

Digital Interpolation

We will study interpolation as it is used in reconstruction of analog signals from its samples

Sampling and Interpolation

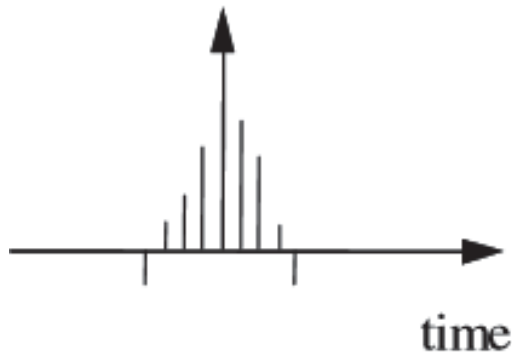
Reconstruction of analog signal from its samples



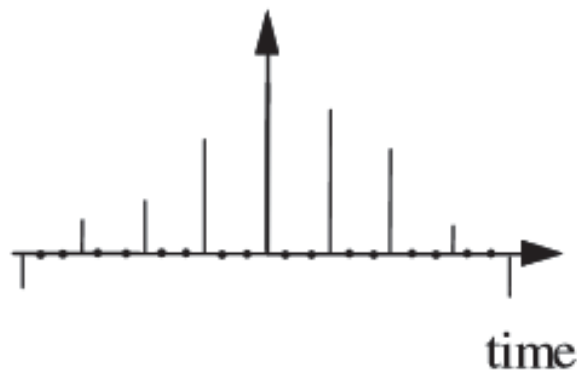
Sampling and Interpolation

Digital Interpolation

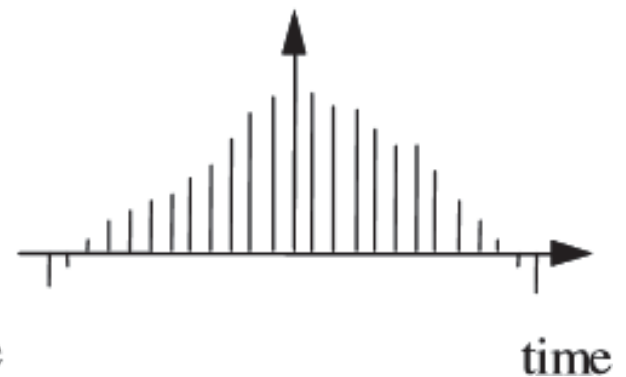
Original signal



Zero inserted signal

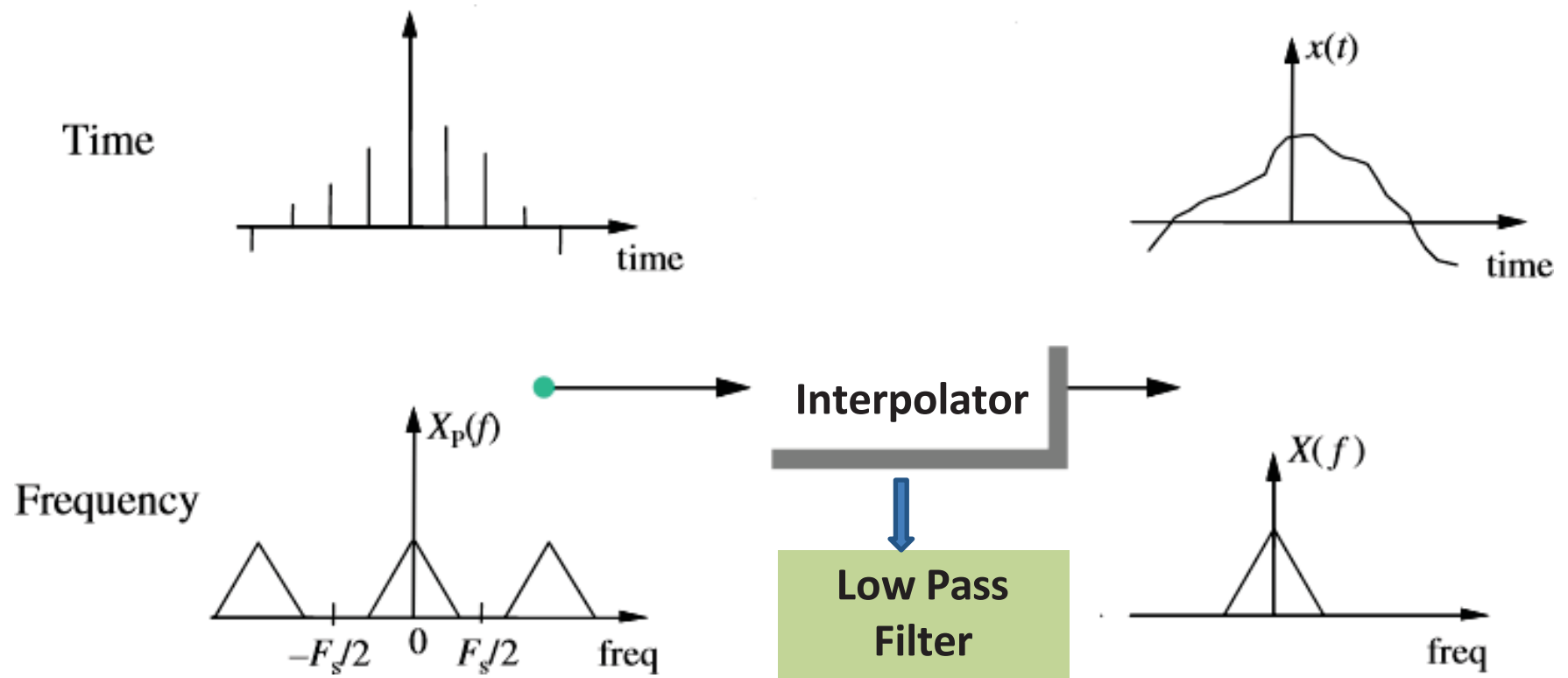


Interpolated signal



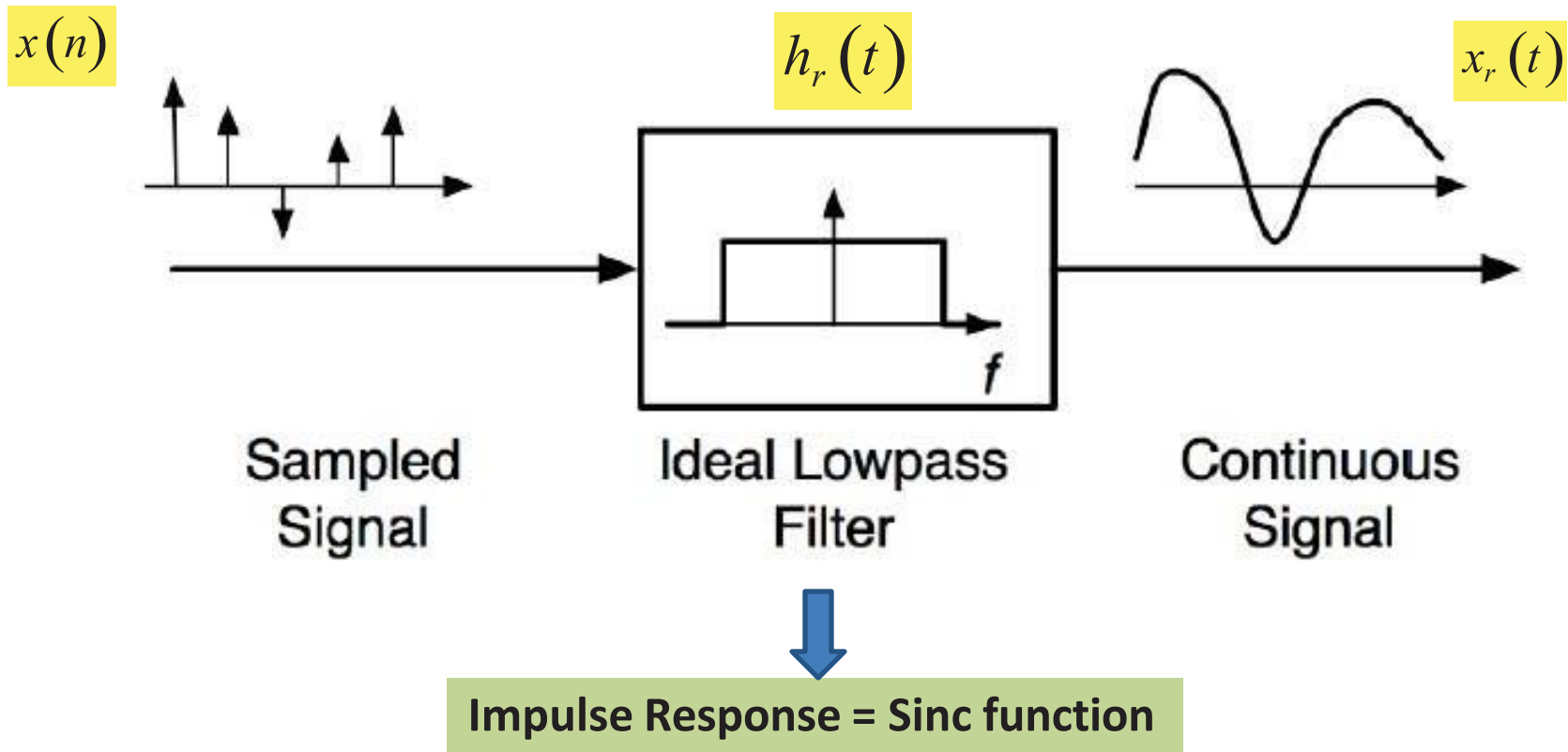
Sampling and Interpolation

Reconstruction of analog signal from its samples



Sampling and Interpolation

Ideal Interpolation: Bandlimited Interpolation



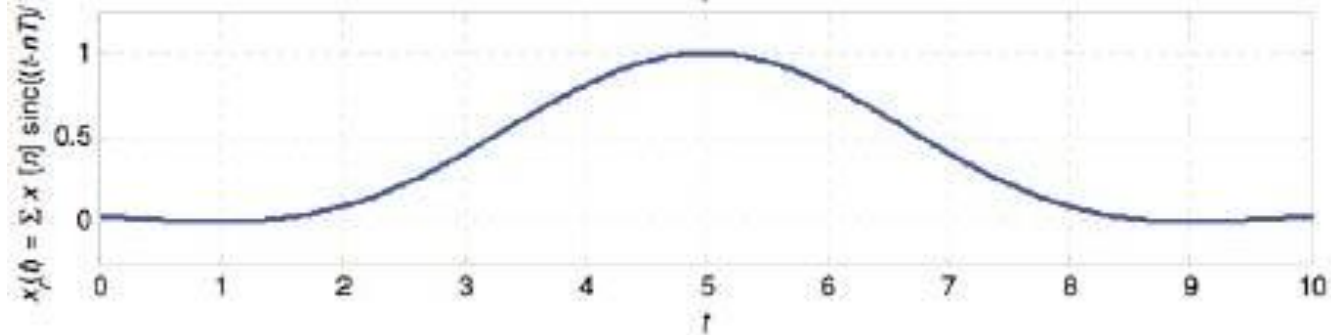
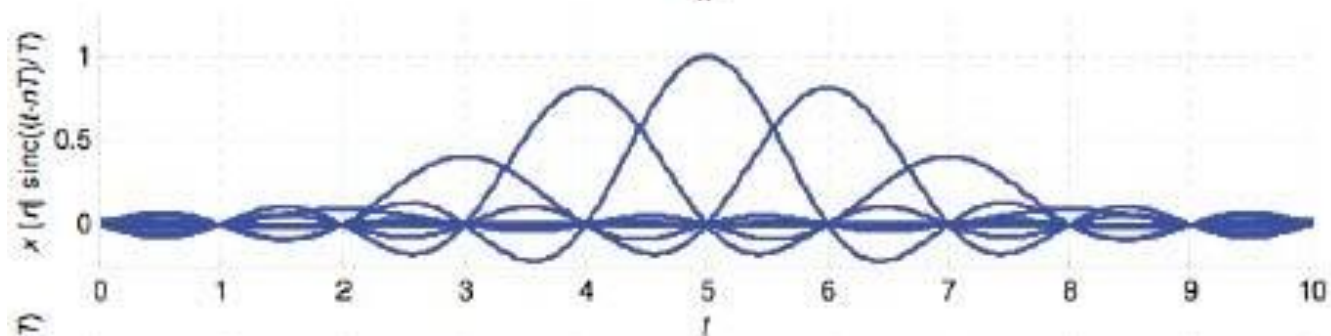
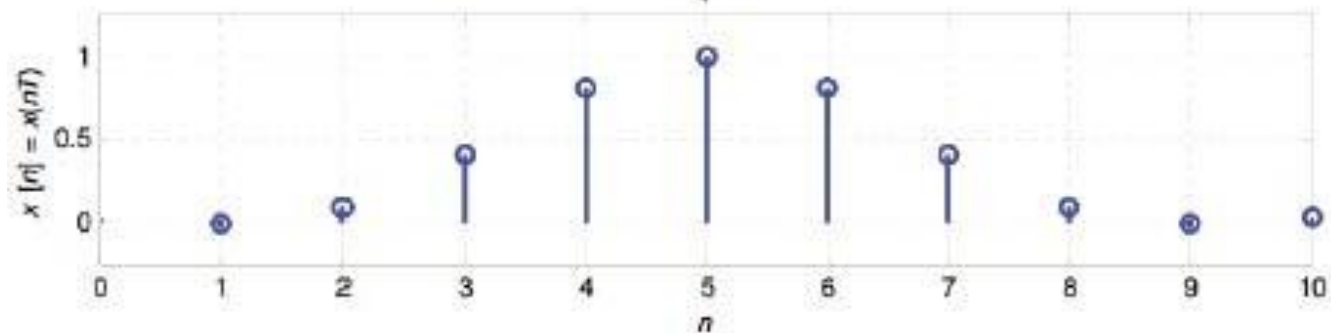
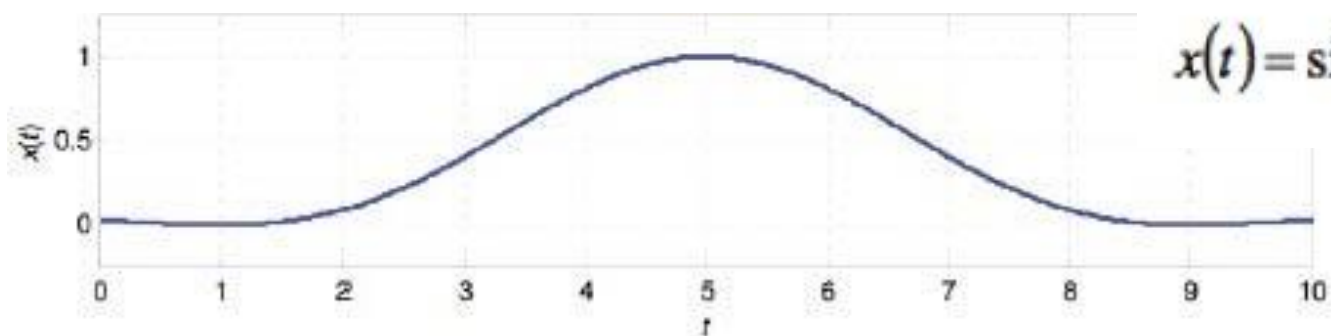
Sampling and Interpolation

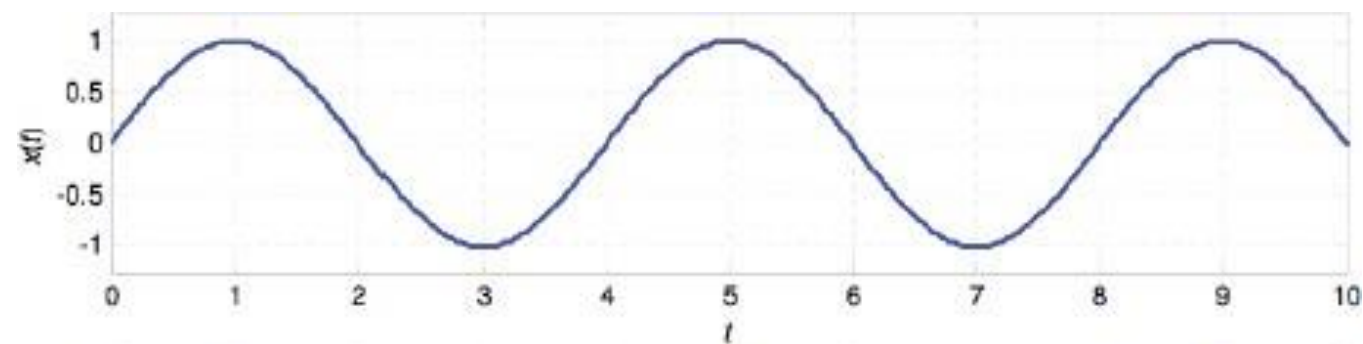
$$\begin{aligned}
 x_r(t) &= x(n) * h(t) = x(nT_s) * h(t) = \left[\sum_{n=-\infty}^{\infty} x(nT_s) \delta(t - nT_s) \right] * h(t) \\
 x_r(t) &= \left[\sum_{n=-\infty}^{\infty} x(nT_s) \delta(t - nT_s) \right] * \left[\frac{\sin(\pi F_s t)}{\pi F_s t} \right] \\
 x_r(t) &= T_s \sum_{n=-\infty}^{\infty} x(nT_s) \left[\frac{\sin(\pi F_s (t - nT_s))}{\pi (t - nT_s)} \right]
 \end{aligned}$$

Recall that an ideal LPF is noncausal and its impulse response is infinitely long! For example, to interpolate $x_r(0.2T_s)$, we need ALL of the samples: $\dots, x(-2T_s), x(-T_s), x(0), \underbrace{x(T_s), x(2T_s), \dots}_{\text{noncausal future data}}$ with different sinc-function weighting coefficients:

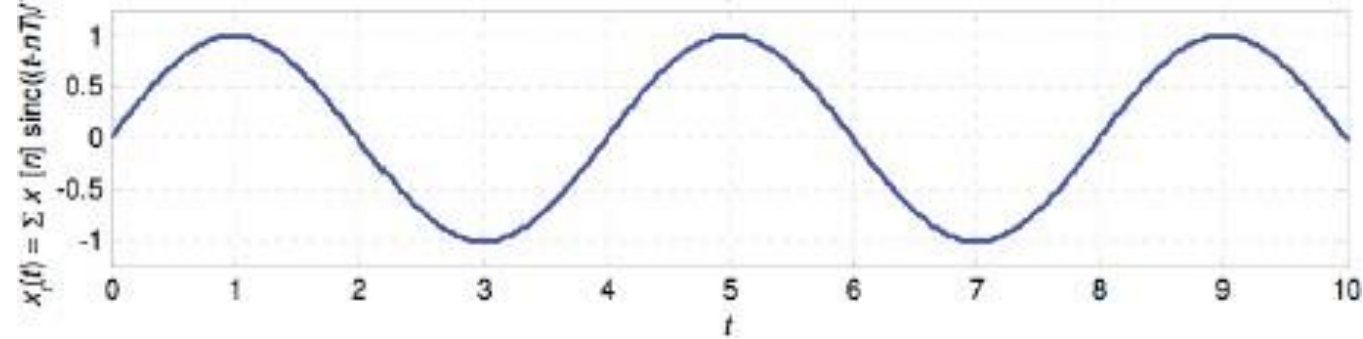
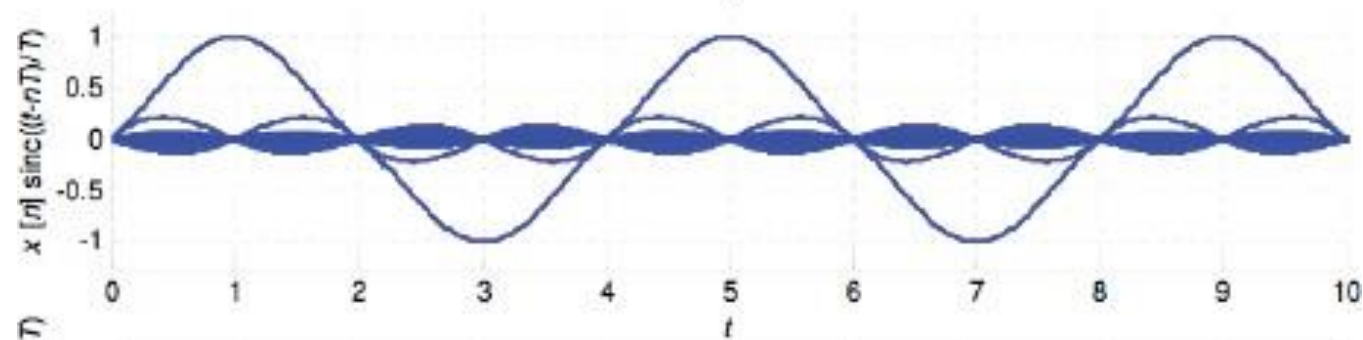
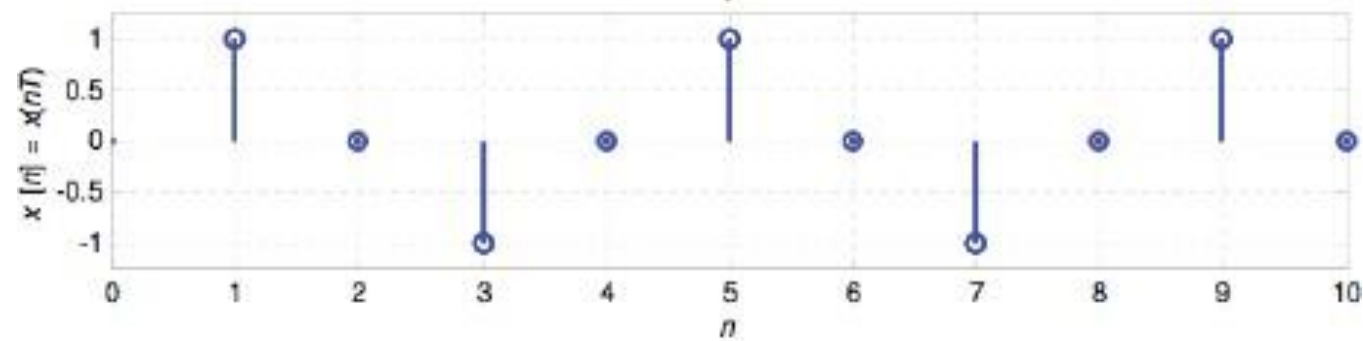
$$\begin{aligned}
 x(0.2T_s) = T_s & \left[\dots + x(-T_s) \frac{\sin \frac{\omega_s}{2}(0.2T_s + T_s)}{\pi(0.2T_s + T_s)} + x(0) \frac{\sin \frac{\omega_s}{2}(0.2T_s - 0)}{\pi(0.2T_s - 0)} \text{ (past inputs)} \right. \\
 & \left. + x(T_s) \frac{\sin \frac{\omega_s}{2}(0.2T_s - T_s)}{\pi(0.2T_s - T_s)} + x(2T_s) \frac{\sin \frac{\omega_s}{2}(0.2T_s - 2T_s)}{\pi(0.2T_s - 2T_s)} + \dots \text{ (future inputs)} \right]
 \end{aligned}$$

$$x(t) = \text{sinc}^2\left(\frac{1}{4}(t-5)\right)$$



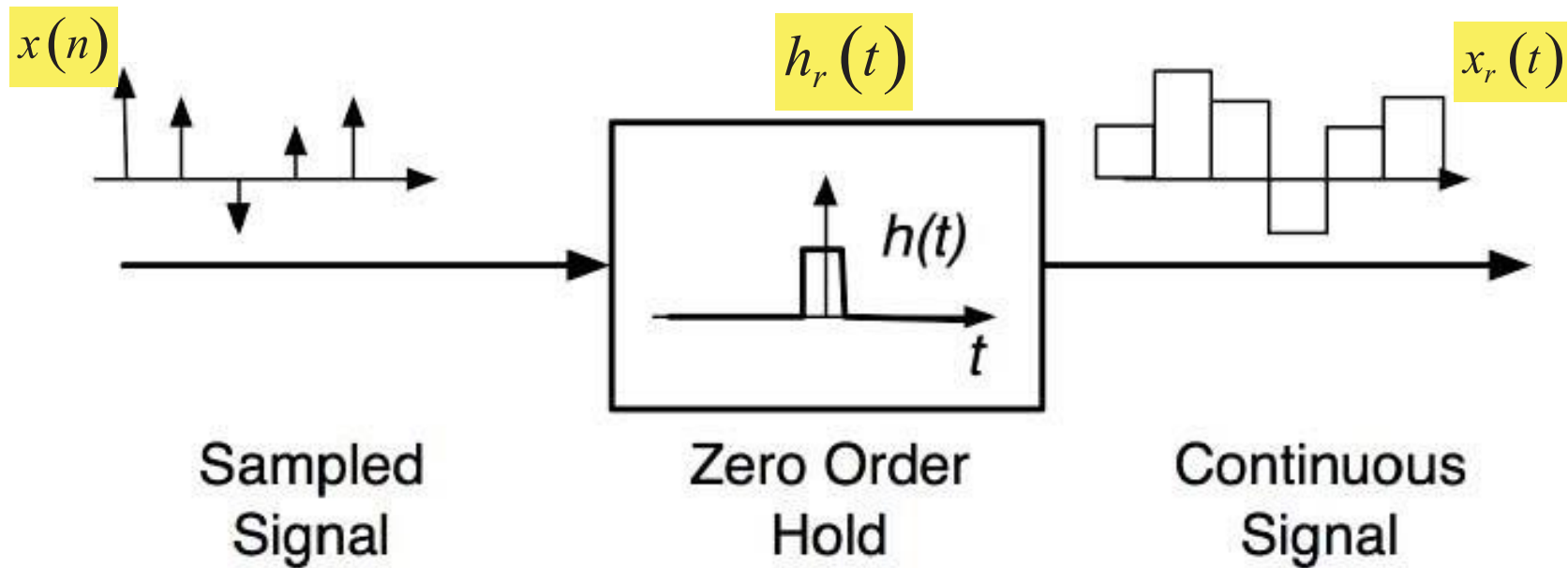


$$x(t) = \sin\left(\frac{\pi}{2}t\right)$$

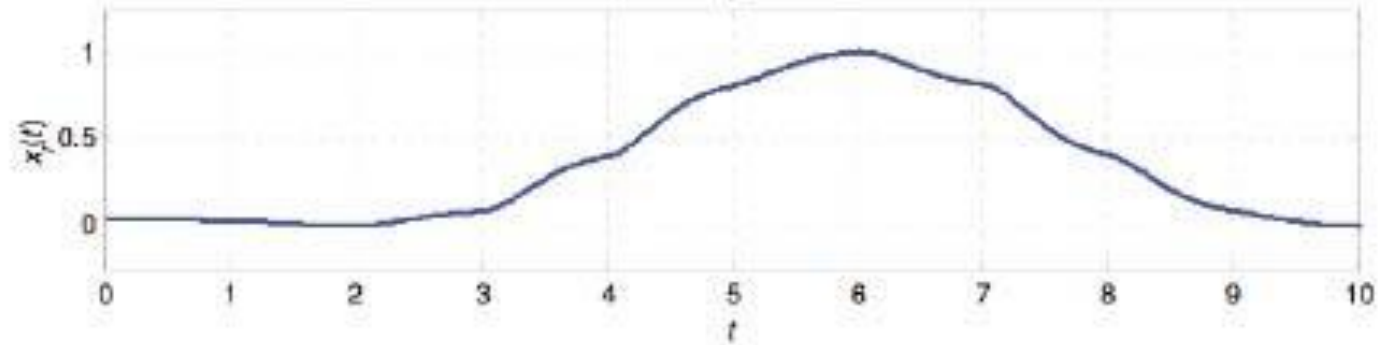
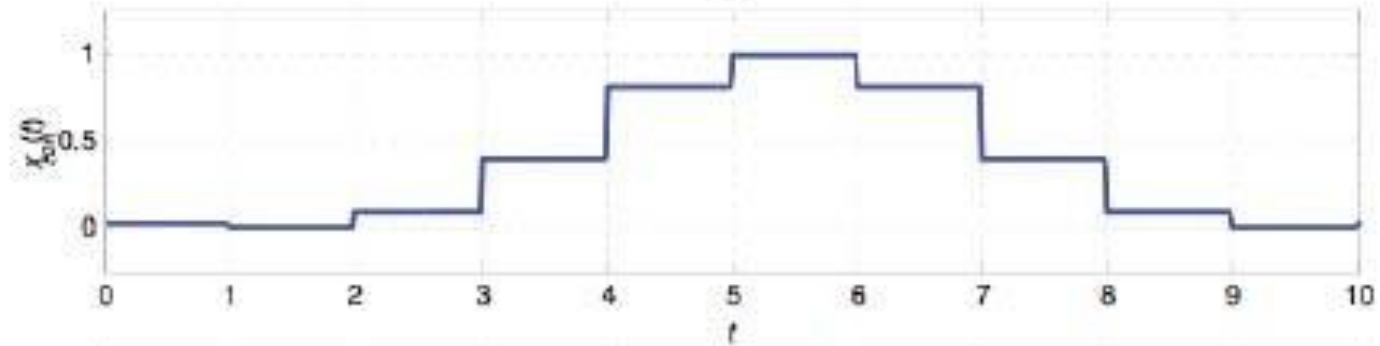
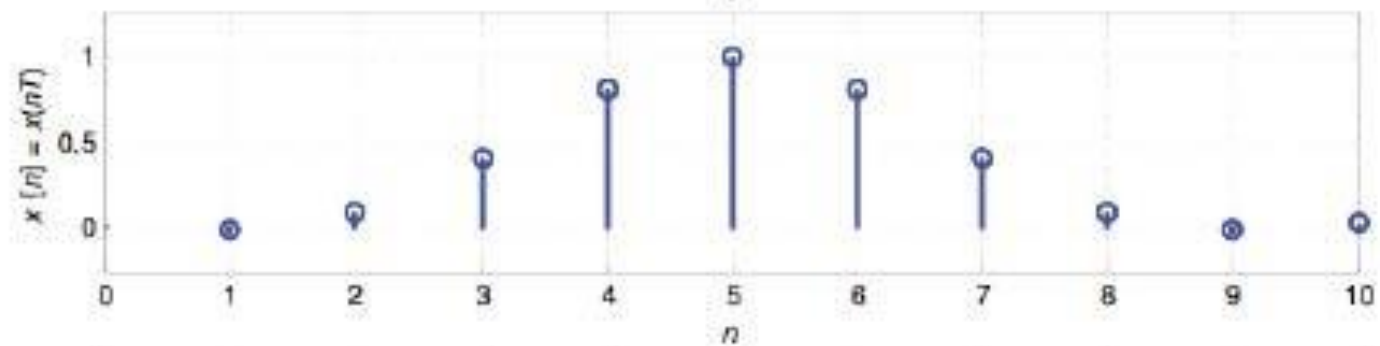
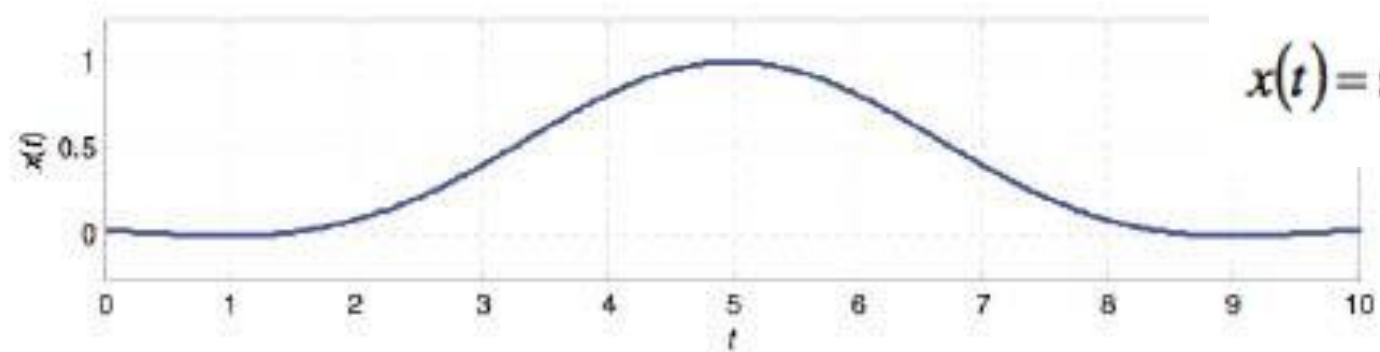


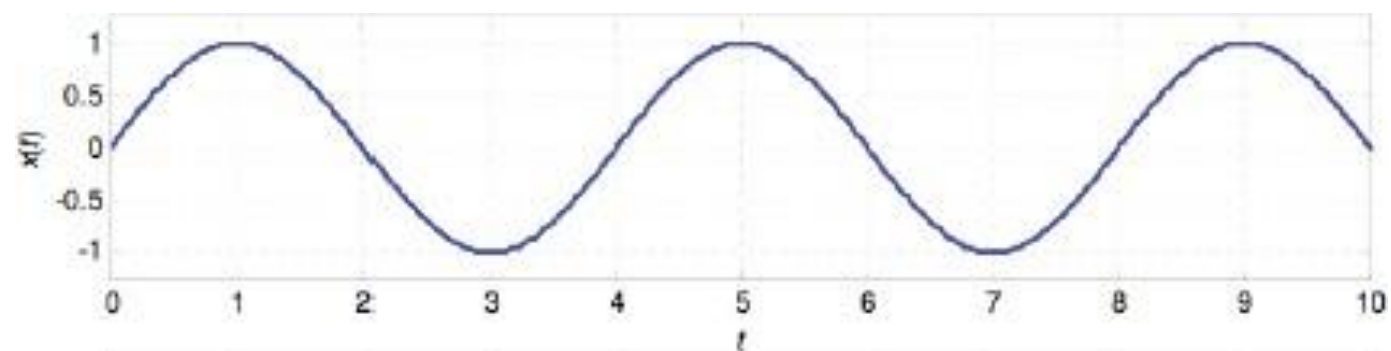
Sampling and Interpolation

Zero order hold

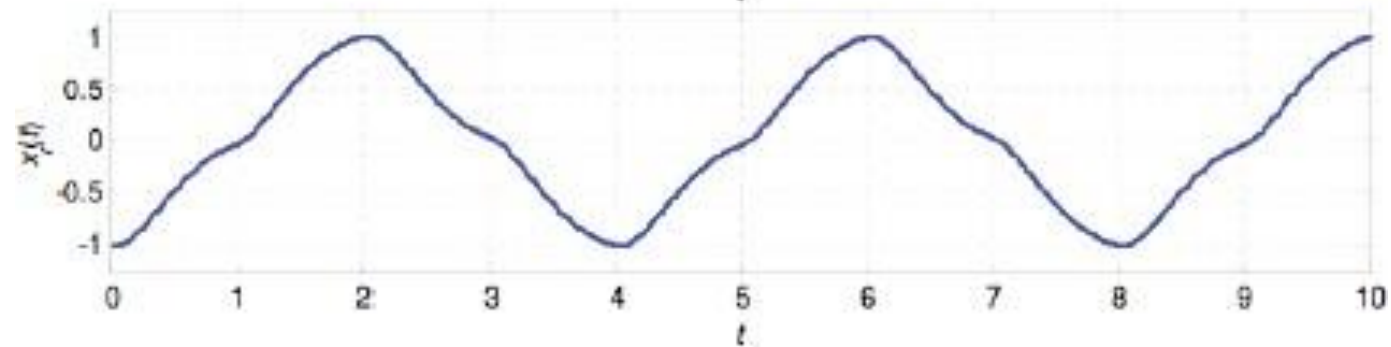
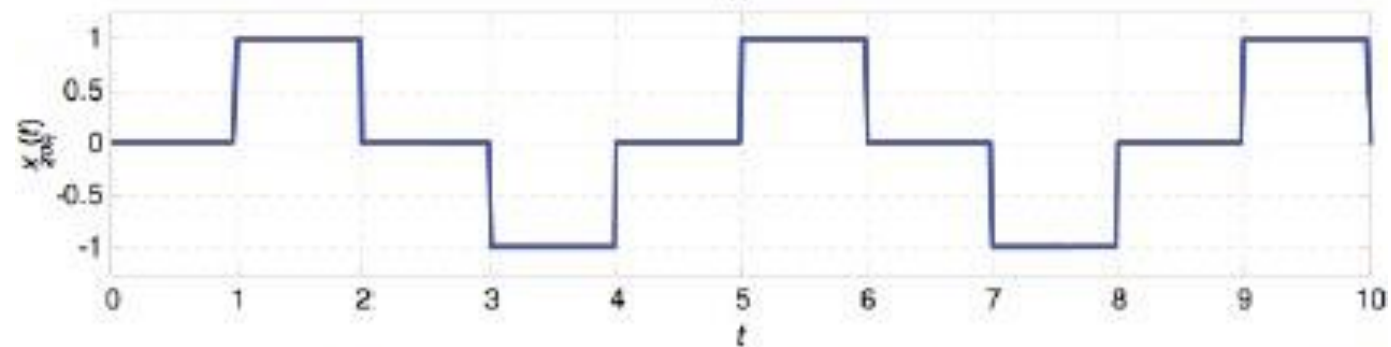
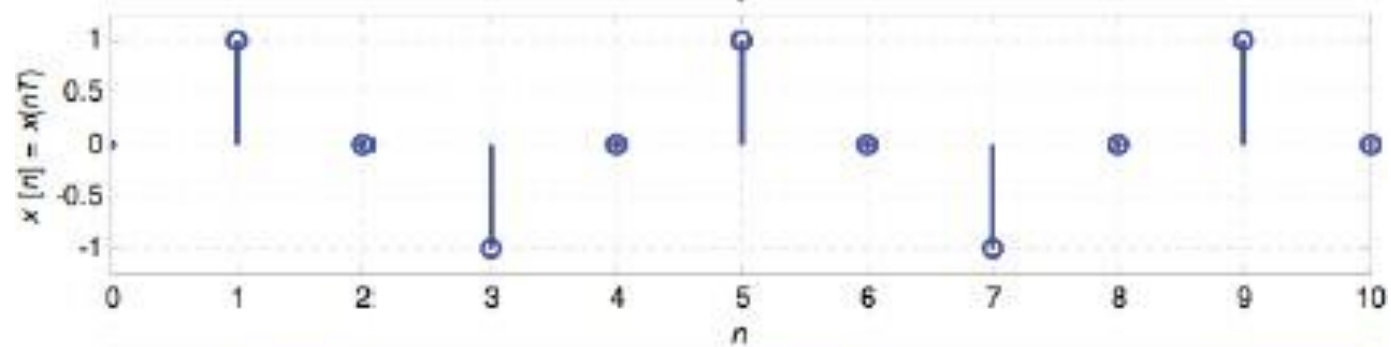


Impulse Response = Rectangular function



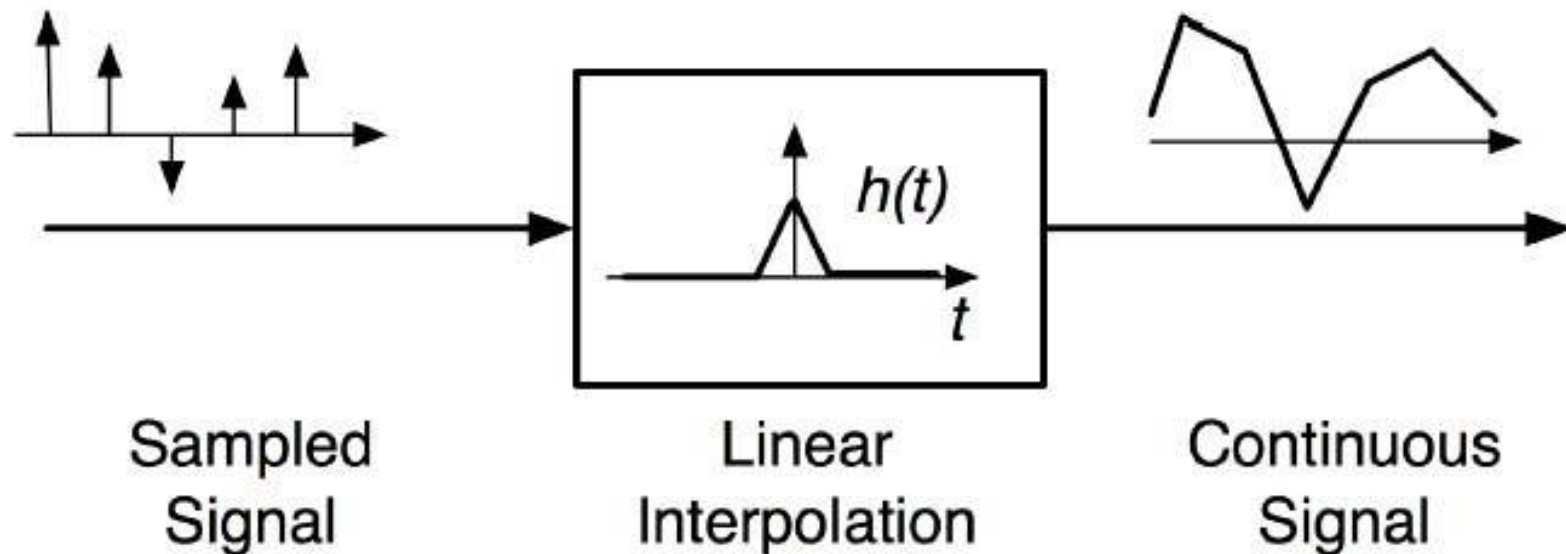


$$x(t) = \sin\left(\frac{\pi}{2}t\right)$$



Sampling and Interpolation

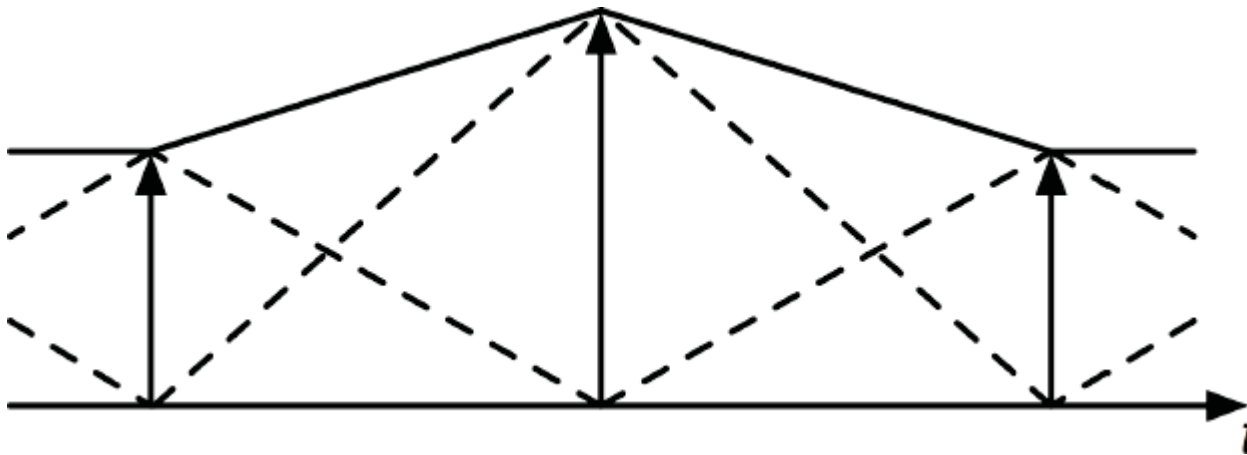
Linear Interpolator: First order hold



Impulse Response = Triangular function

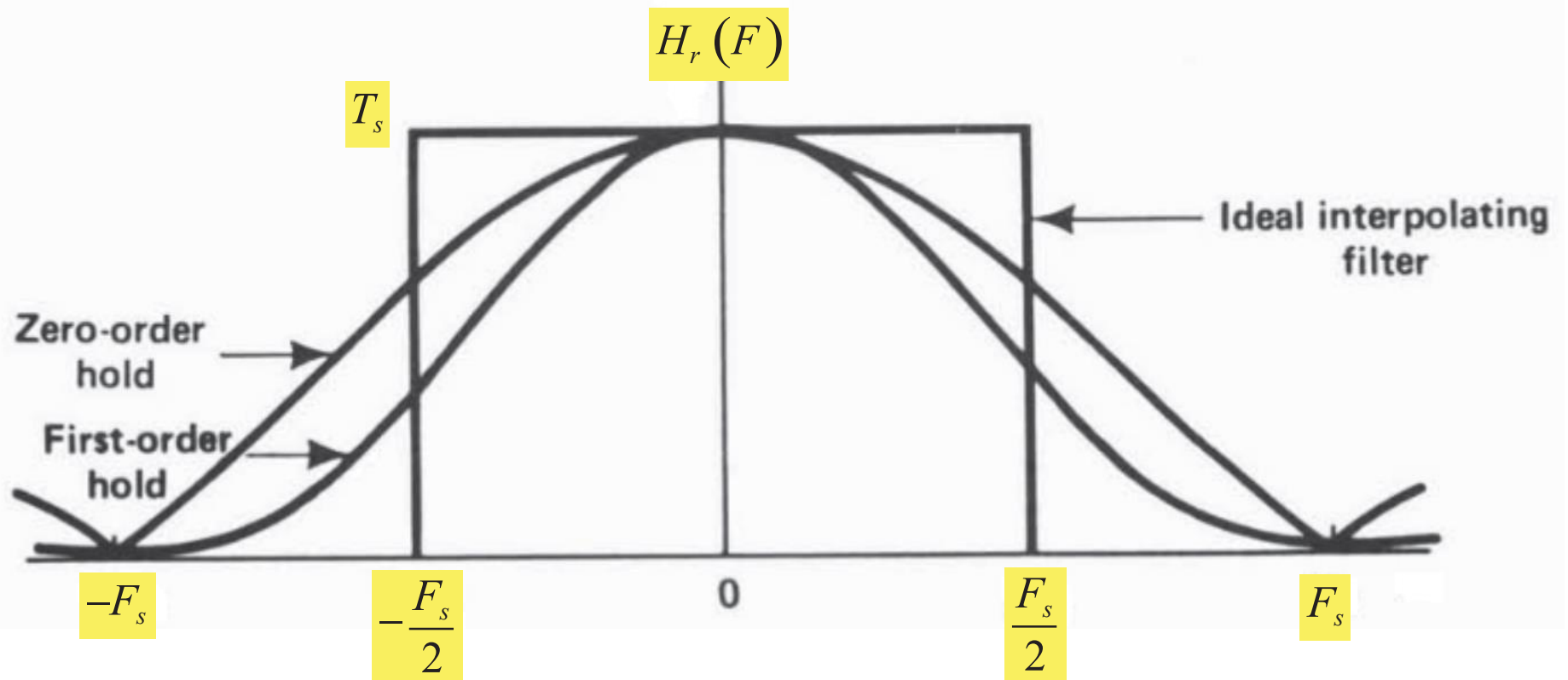
Sampling and Interpolation

Linear Interpolator: First order hold



Sampling and Interpolation

Reconstruction Filters in Frequency Domain



Outcomes:

- Design of anti-aliasing filter: given sampling rate, what will be its cut-off frequencies and its impulse response
- Spectrum of signal before and after sampling
- Spectrum of sampling function e.g. ideal impulse train.
- Design of reconstruction/interpolation filter: its cut-off frequencies and impulse response.
- Spectrum of signal after reconstruction.